



INTRODUCTION TO DATA COMMUNICATION AND COMPUTER NETWORK (IDCCN) **[CSIT369]**

MODULE II

Basic Concepts of Network

(UG - III Sem)

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COMPUTER NETWORK

➤ computer network is a collection of interconnected computing devices that communicate and share resources using communication links and protocols.

➤ Why Networks?

- Resource Sharing
- Internet Access
- Communication

➤ Examples

- Home Wi-Fi Network
- Mobile Network
- Internet

➤ Real-Life Applications

- WhatsApp Messaging
- YouTube Streaming
- Online Banking



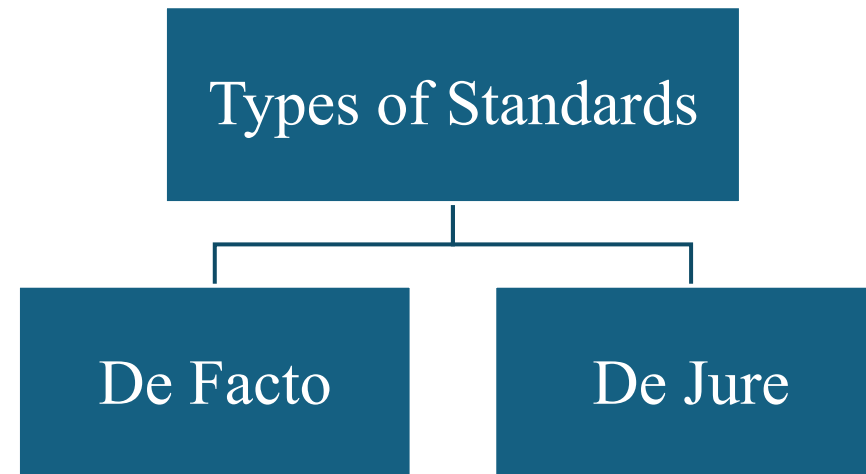


STANDARDS AND ORGANIZATIONS

“Network standards are agreed-upon rules and specifications that define how data is transmitted, received, and processed in a network, ensuring interoperability and reliable communication among devices from different manufacturers.”

- **Why Standards?**

- Imagine if every company used different networking rules? Communication would fail.
- Ensure compatibility among devices.
- Enable communication between different vendors.
- Promote interoperability.





DE FACTO STANDARDS

*“De facto (Latin for "**from the fact**") standards emerge through widespread use and acceptance rather than through formal approval by a standards organization.”*

Characteristics:

- Developed through common practice and market adoption.
- Become standards because they are widely used.
- May later be formalized by standardization bodies.

Examples:

- **HTTP:** Initially developed by Tim Berners-Lee at CERN and became the foundation of the World Wide Web.
- **Bluetooth:** Originally developed by Ericsson and later adopted worldwide for wireless communication.



DE JURE STANDARDS

“De jure (Latin for "by law") standards are formally approved and established by recognized standardization organizations.”

Characteristics:

- Developed through a structured and official process.
- Ensure global compatibility and interoperability.
- Widely accepted across industries and countries.



INTERNATIONAL STANDARDIZATION ORGANIZATIONS

1. Treaty-Based Organizations

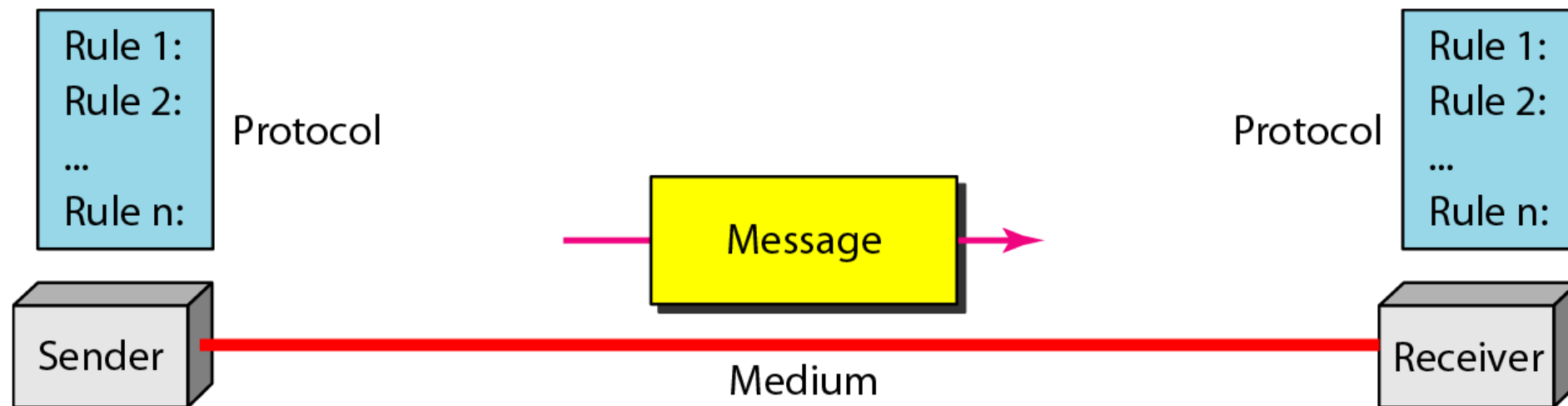
- Established through agreements among national governments.
- **ITU (International Telecommunication Union)**
Develops global telecommunications standards.
- **ISO (International Organization for Standardization)**
Creates international standards across various technological fields.

2. Voluntary Organizations

- Independent organizations formed through voluntary participation.
- **IETF (Internet Engineering Task Force)**
Develops Internet protocols such as TCP/IP.
- **IEEE (Institute of Electrical and Electronics Engineers)**
Develops networking standards such as Ethernet (IEEE 802.3) and Wi-Fi (IEEE 802.11).

Introduction to Data Communication

- The term telecommunication means communication at a distance.
- The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.



Components of a data communication system



Five Components of Data Communication:

- Message (data) to be communicated
- Sender
- Receiver
- Transmission medium Physical path by which a message travels
- Protocol A set of rules that govern data communication



Networks:

- A network is a set of devices (often referred to as nodes) connected by communication links.
- A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.
- A link can be a cable, air, optical fiber, or any medium which can transport a signal carrying information.

Network Criteria:

- ❖ **Performance**
 - ❖ **Depends on Network Elements**
 - ❖ **Measured in terms of Delay and Throughput**
- ❖ **Reliability**
 - ❖ **Failure rate of network components**
 - ❖ **Measured in terms of availability/robustness**
- ❖ **Security**
 - ❖ **Data protection against corruption/loss of data due to:**
 - ❖ **Errors**
 - ❖ **Malicious users**



Protocol:

- A protocol is synonymous with rule.
- It consists of a set of rules that govern data communications.
- It determines what is communicated, how it is communicated and when it is communicated.
- The key elements of a protocol are syntax, semantics and timing.



Elements of a Protocol:

- ▶ Syntax
 - ▶ Structure or format of the data
 - ▶ Indicates how to read the bits
- ▶ Semantics
 - ▶ Interprets the meaning of the bits
 - ▶ Knows which fields define what action
- ▶ Timing
 - ▶ When data should be sent and what
 - ▶ Speed at which data should be sent or speed at which it is being received.



Advantages

1. Resource Sharing: The aim is to make all programs, data and peripherals available to anyone on the network irrespective of the physical location of the resources and the user.
2. Reliability: A file can have copies on two or three different machines, so if one of them is unavailable (hardware crash), the other copies could be used. For military, banking, air reservation and many other applications it is of great importance.
3. Cost Factor: Personal computers have better price / performance ratio than micro computers. So, it is better to have PC's, one per user, with data stored on one shared file server machine.
4. Communication Medium: Using a network, it is possible for managers, working far apart, to prepare financial report of the company changes at one end can be immediately noticed at another and hence it speeds up co-operation among them.
5. Central Storage of Data: Files can be stored on a central node (the file server) that can be shared and made available to every user in an organization.
6. Anyone can connect to a computer network: There is a negligible range of abilities required to connect to a modern computer network.
7. Computer networks allow us to be more productive with our time.
8. A computer network can share a single Internet connection.
9. It provides people with an opportunity to solve problems creatively.



Disadvantages

- Cost of network:

The expense of executing the system including cabling, fitting, installation and equipment can be expensive.

- Virus and Malware:

On the off chance that even one PC on a system gets contaminated with an infection, there is a possibility for alternate frameworks to get tainted as well. Infections can spread on a system effectively, in view of the between availability of different gadgets.

- Needs an Efficient Handler:

Managing a big network is complex which requires training and generally a network manager needs to be hired. The technical competences and expertise needed to operate and oversee a computer network is substantially high.

- Lack of Robustness:

If the core file server of a computer network fails, the entire system may become useless. If there is a failure in the main connecting server or a bridging device in the network, the entire network will come to a halt.



Applications of Computer Networking

- Telephone communication
- Accessing Remote databases
- Accessing Remote programs
- Value added communication facility
- Marketing and sales
- Financial services
- Manufacturing company
- Electronic message
- Teleconferencing
- Cable television



Line Configuration

A network is two or more devices connected through a link. A link is a communication pathway that transfer data from one device to another. Devices can be a computer, printer or any other device that is capable to send and receive data.

For visualization purpose, imagine any link as a line drawn between two points.

For communication to occur, two devices must be connected in some way to the same link at the same time.

There are two possible types of connections:

- 1. Point-to-Point Connection**
- 2. Multipoint Connection**



1.Point-to-PointConnection:

1. The entire capacity of the link is reserved for transmission between those two devices:
2. A point-to-point connection provides a dedicated link between two devices.
3. Most point-to-point connections use an actual length of wire or cable to connect the two end, but other options such as microwave or satellite links are also possible.
4. Point to point network topology is one of the easiest and most conventional network topologies.
5. It is also the simplest to establish and understand.

Example: Point-to-Point connection between remote control and Television for changing the channels.

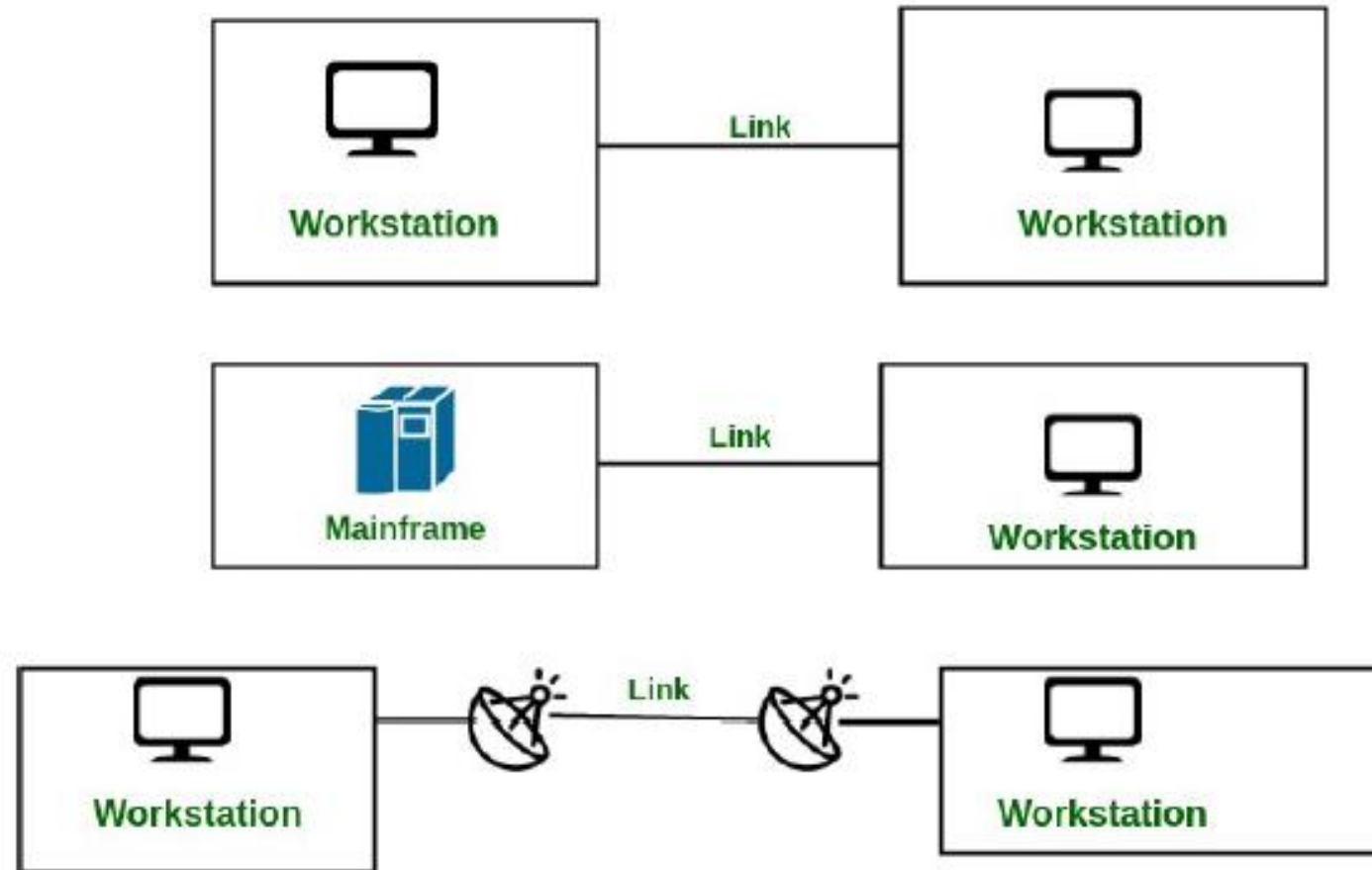


Figure: Point-to-Point Connection



2. Multipoint Connection:

1. In this connection, two or more devices share a single link. It is also called Multidrop configuration.
2. More than two devices share the link that is the capacity of the channel is shared now. With shared capacity, there can be two possibilities in a Multipoint Line configuration:

Spatial Sharing: If several devices can share the link **simultaneously**, it called Spatially shared line configuration.

Temporal (Time) Sharing: If users must take turns using the link , then it called Temporally shared or **Time-Shared** Line configuration.

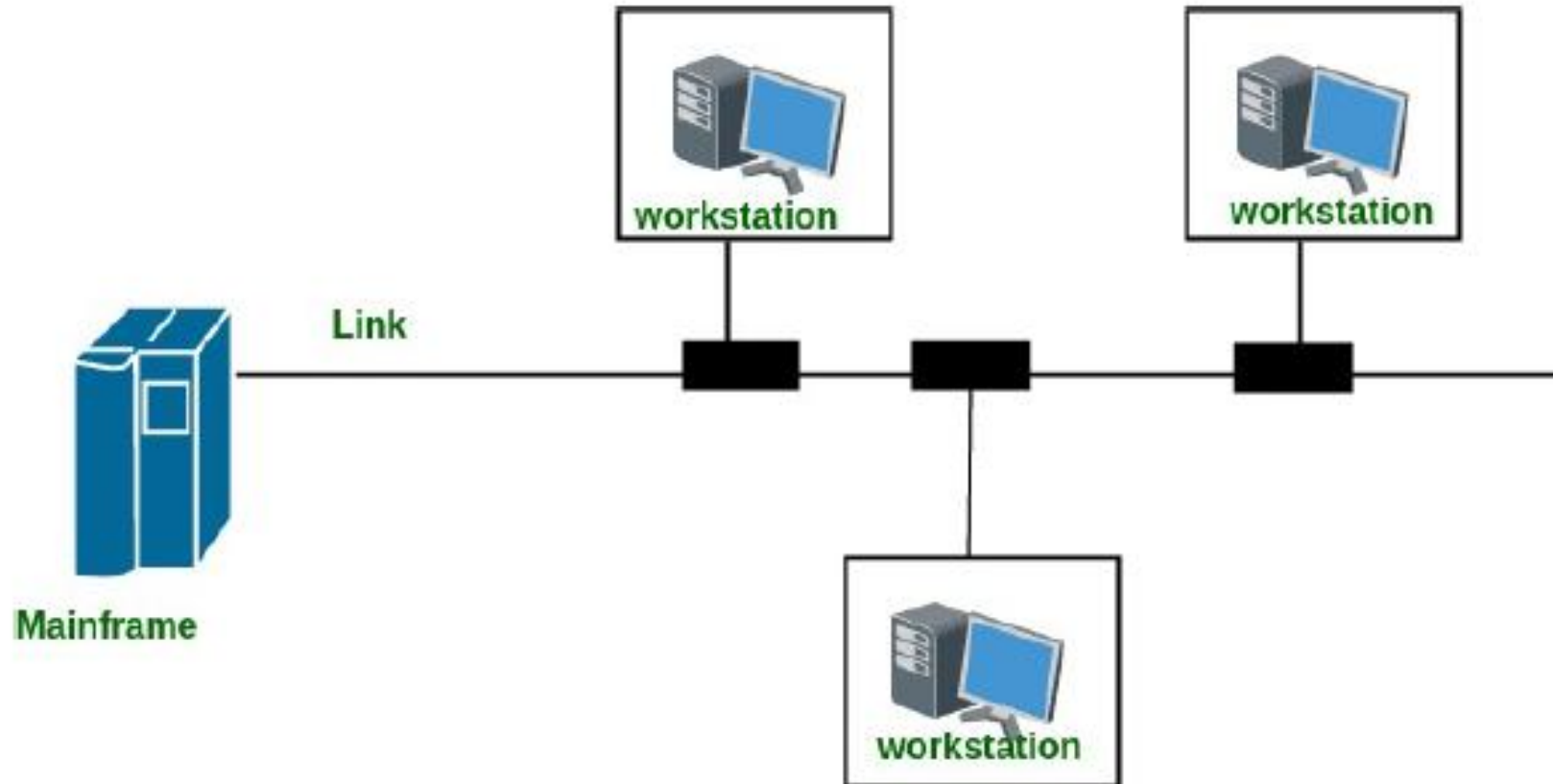


Figure: Multipoint Connection

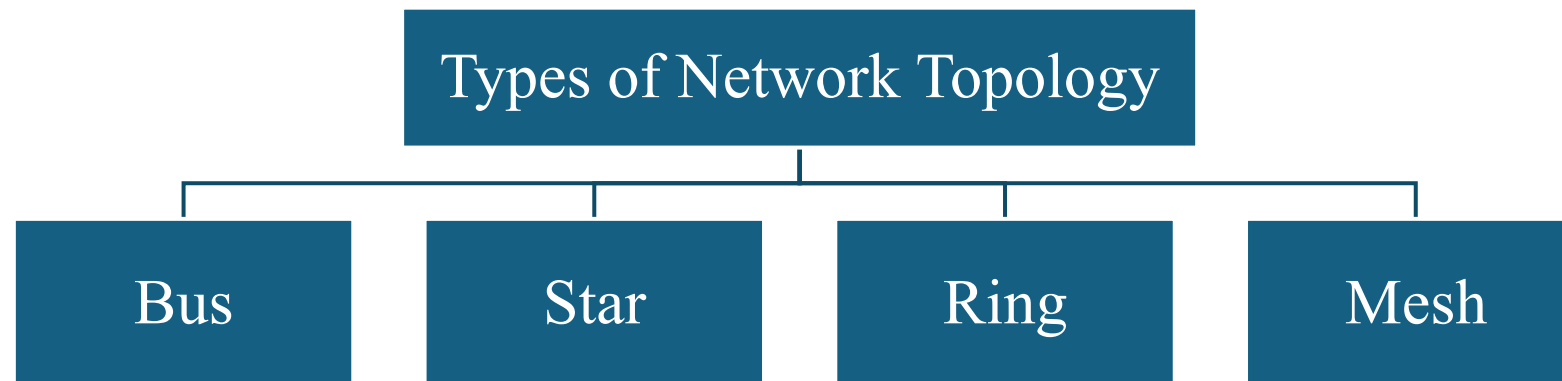


NETWORK TOPOLOGY

“Network topology refers to the physical or logical arrangement of nodes (computers, switches, routers, etc.) and communication links in a network.”

Importance

- Determines network performance.
- Affects reliability and scalability.
- Influences installation and maintenance costs.





Topology:

- A network topology is the arrangement of a network, including its nodes and connecting lines.
- There are two ways of defining network geometry: the physical topology and the logical (or signal) topology.
- Physical topology of networks refers to the configuration of cables, computers and other peripheral devices
- Logical (or signal) topology refers to the nature of the paths the signals follow from node to node.
- In many instances, the logical topology is the same as the physical topology. But this is not always the case.
- For example, some networks are physically laid out in a star configuration, but they operate logically as bus or ring networks.



Topology Contd....:

- When design a network, choice of topology will be determined by the size, architecture, cost and management of network.
- Basic types of network topologies includes:
 - Bus Topology
 - Star Topology
 - Ring Topology
 - Mesh Topology



BUS TOPOLOGY

“In a Bus Topology, all devices are connected to a single communication cable called the **backbone**.”

Advantages:

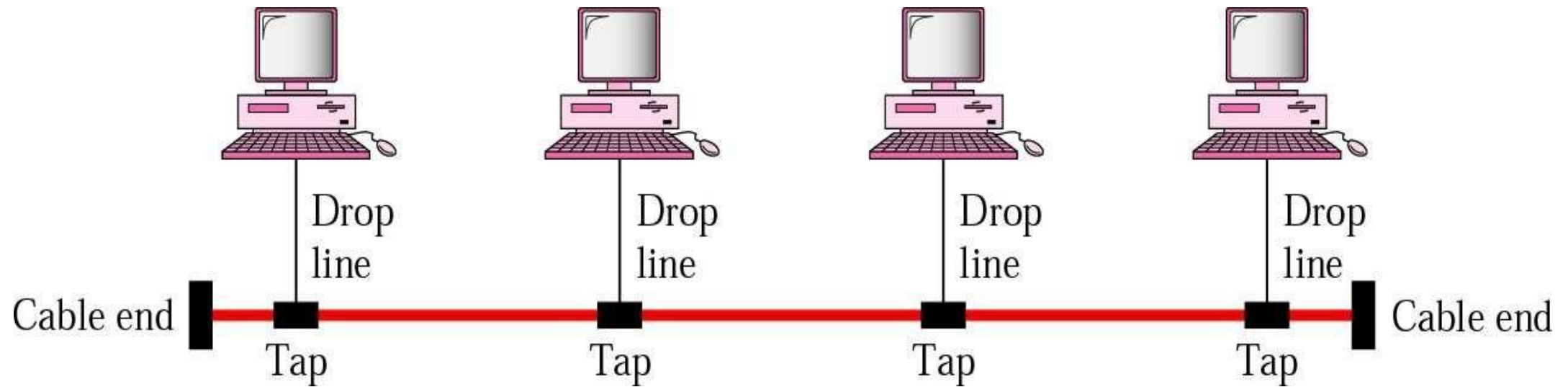
- Easy installation
- Low cost
- Requires less cable

Disadvantages:

- Backbone failure affects entire network
- Difficult fault isolation
- Performance decreases with traffic



BUS TOPOLOGY





BUS TOPOLOGY

- It is used usually when a network installation is small, simple or temporary.
- Bus topology is a network type in which every computer and network device is connected to single cable.
- When one computer sends a signal up the cable, all computers on the network receives the information but one with the address that matches the one encoded information while all others rejects the message.
- Speed is slow as only one computer can send a message at a time.
- A computer must wait until bus topology requires proper termination at both ends of the cable.

STAR TOPOLOGY

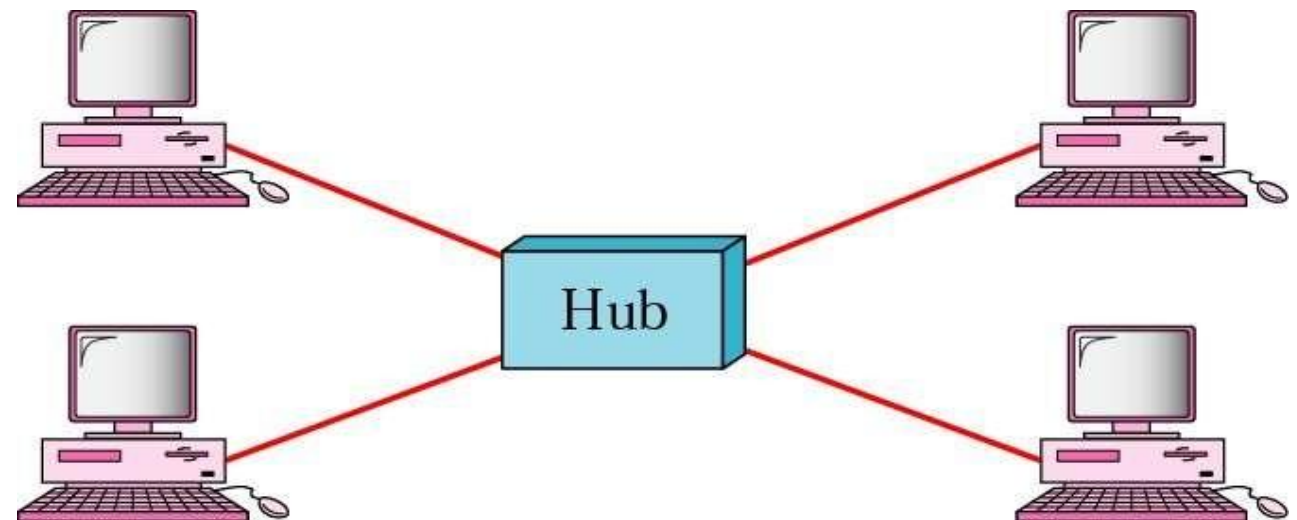
“All devices are connected to a central device such as a switch or hub.”

Advantages:

- Easy management
- Easy fault detection
- Highly reliable

Disadvantages:

- Central device failure affects network
- More cable required





STAR TOPOLOGY

- Links computers by individual cables to a central unit usually a hub.
- When a computer component transmits a signal to the network, the signal travels to the hub.
- Then hub broadcast signal to all other components connected to the hub.
- It can be expanded by placing another star hub.



RING TOPOLOGY

“Each device is connected to two neighboring devices, forming a closed loop.”

Advantages:

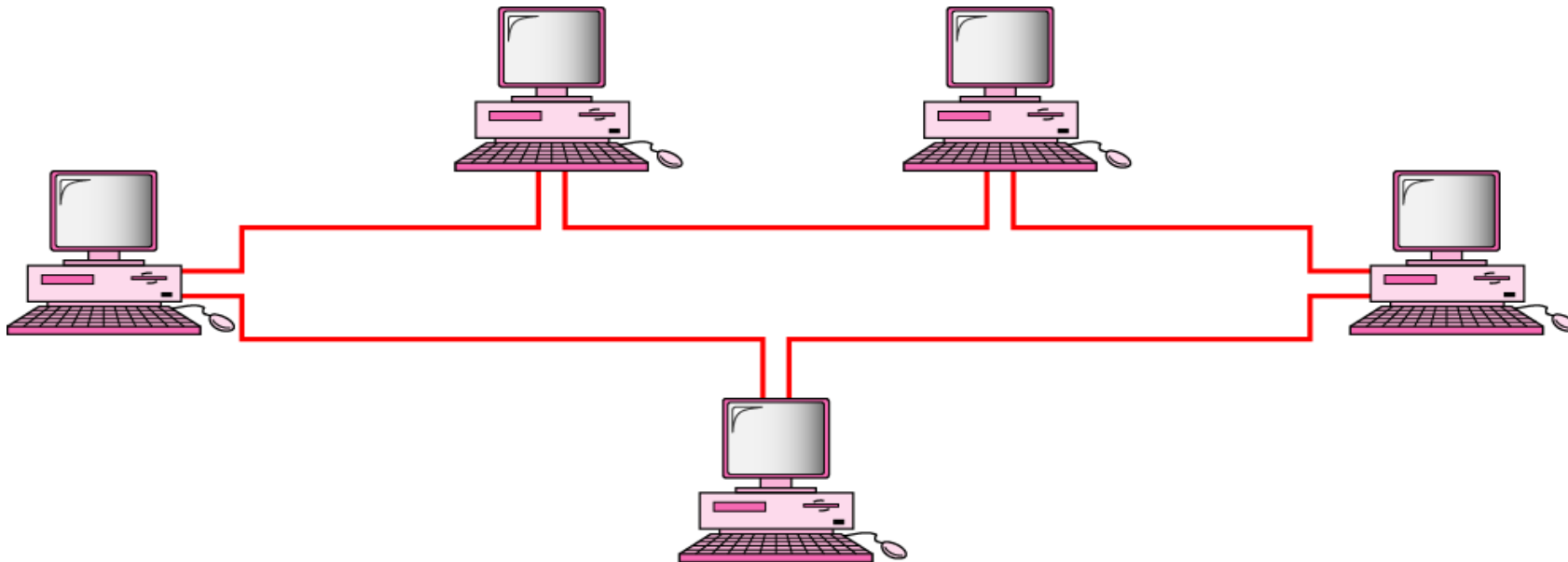
- Orderly data transmission
- No data collision

Disadvantages:

- Single node failure can disrupt communication
- Difficult troubleshooting



RING TOPOLOGY





RING TOPOLOGY

- Rings are used in high performance networks where large bandwidth is necessary.
- Message flow around the ring in one direction.
- Some ring network do ring token passing.
- A short message called token (memory area) is passed around a ring until a computer wishes to send information to other computers.
- That computer modifies token, adds an electronic address and data and send it around the ring.
- Each computer in sequence receives the token and next computer until either the electronic address matches the address of a computer Or the token return to its origin .
- The receiving computer returns a message to the sender indicating that message has been received.

MESH TOPOLOGY

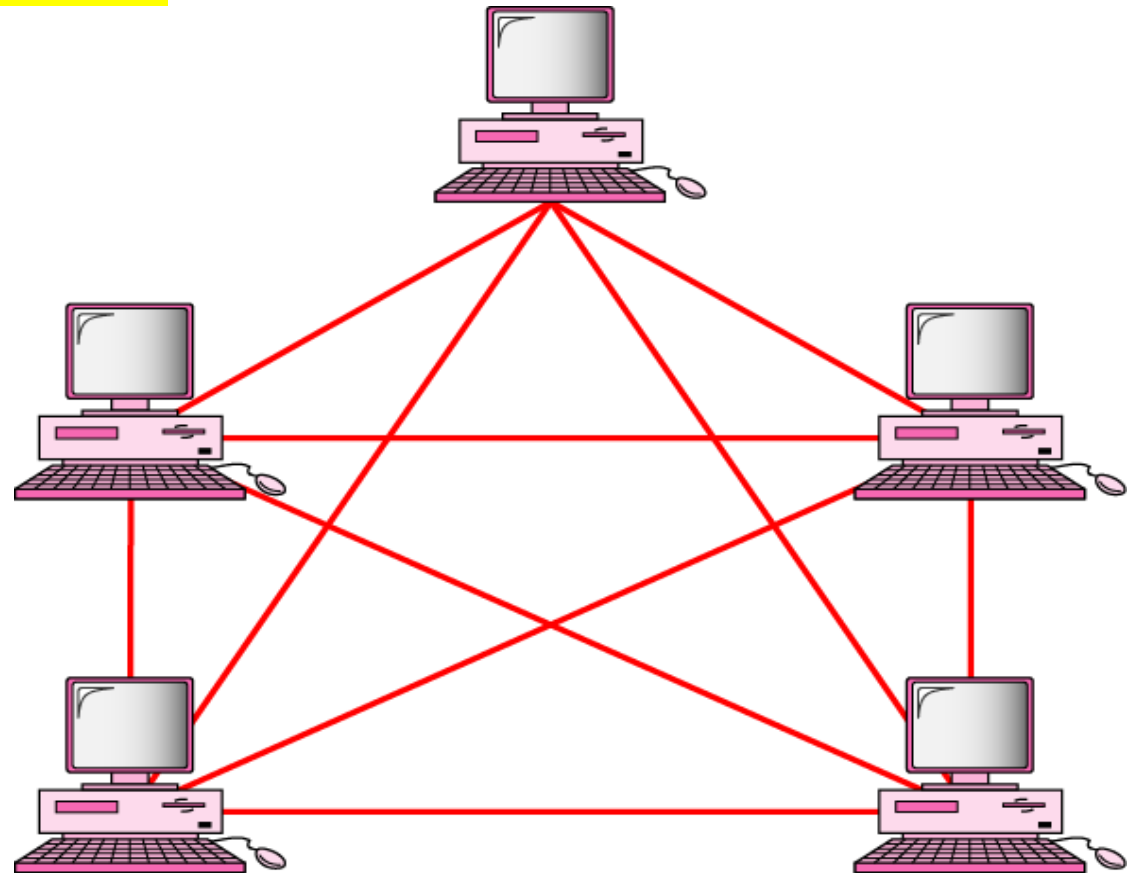
“Every device is connected directly to every other device.”

Advantages:

- High reliability
- Fault tolerant
- Secure communication

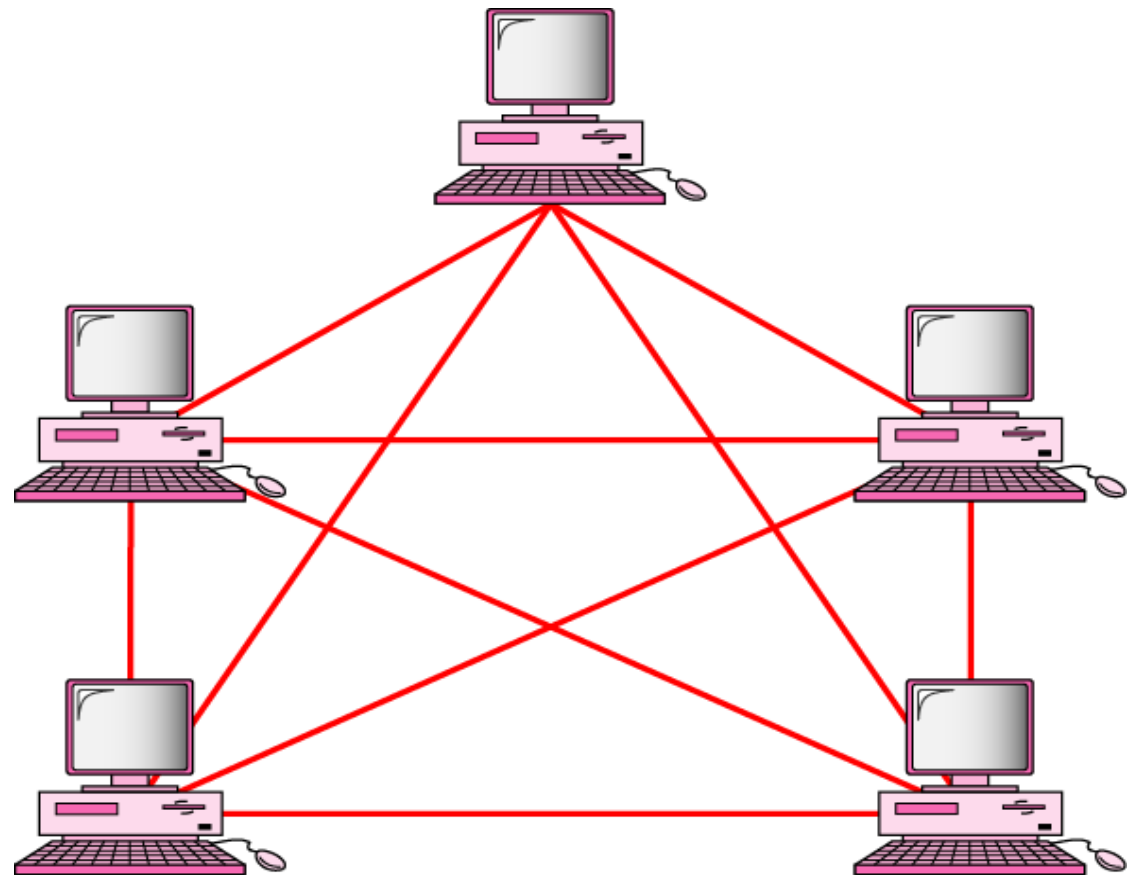
Disadvantages:

- Expensive
- Complex installation



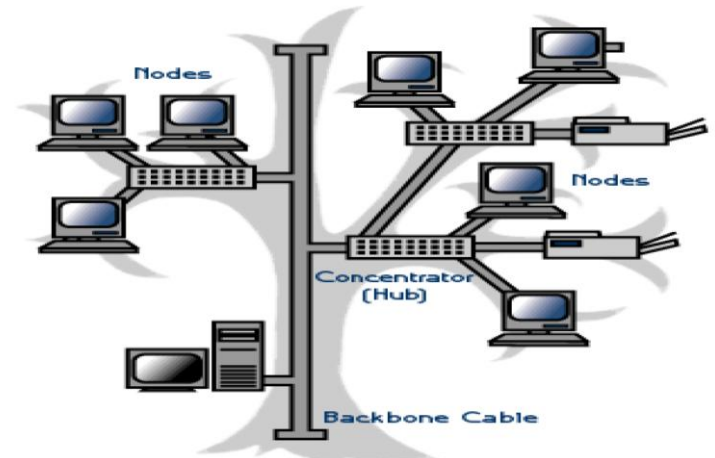
MESS TOPOLOGY

- In a mesh topology each computer on network has redundant data paths.
- The mesh topology provides fault tolerance – if a wire, hub, switch or other components fails, data can travel along an alternate path.



TREE TOPOLOGY

- A tree topology combines characteristics of linear bus and star topologies.
- It consists of groups of star-configured workstations connected to a linear bus backbone cable.
- Tree topologies allow for the expansion of an existing network and enable us to configure a network to meet their needs.



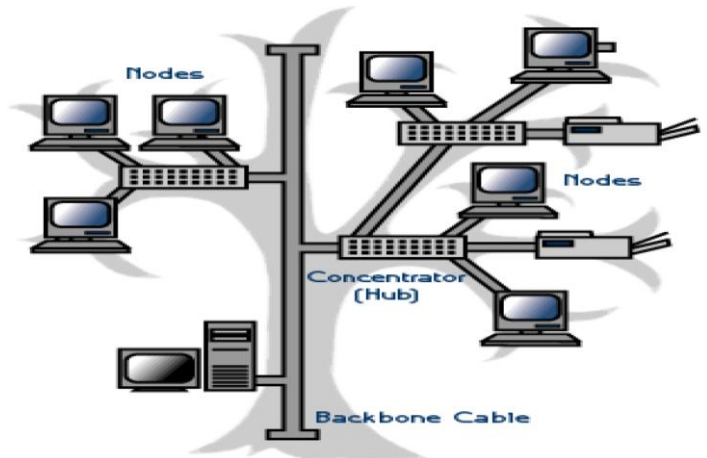
TREE TOPOLOGY

- **Advantage:**

- it is easier to add new nodes or branches to the network.
- A Tree topology is a good choice for large computer networks as the tree topology "divides" the whole network into parts that are more easily manageable.

- **Disadvantage:**

- Cabling cost is more.
- The entire network depends on a central hub and a failure of the central hub can cripple the whole network.





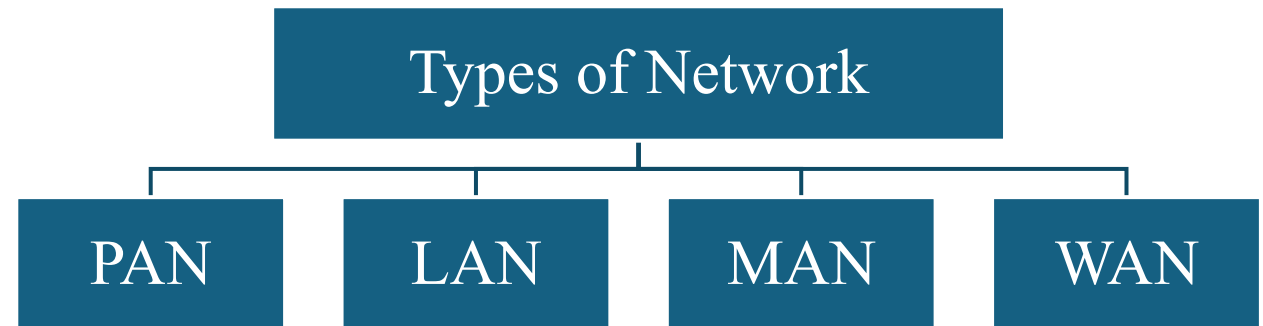
NETWORK CLASSIFICATIONS

“Network can be understood as the interconnection of computers.”

“Network classification is the process of categorizing computer networks based on their geographical coverage area.”

Types of Networks

1. Personal Area Network (PAN)
2. Local Area Network (LAN)
3. Metropolitan Area Network (MAN)
4. Wide Area Network (WAN)



“As coverage increases, complexity and cost generally increase.”

PERSONAL AREA NETWORK (PAN)

“A PAN is a network that connects devices within an individual's personal workspace.”

Characteristics

- Coverage: Up to 10 meters
- Low cost
- Usually wireless

Examples

- Bluetooth headset
- Smartwatch
- Wireless keyboard and mouse

PERSONAL AREA NETWORK



LOCAL AREA NETWORK (LAN)

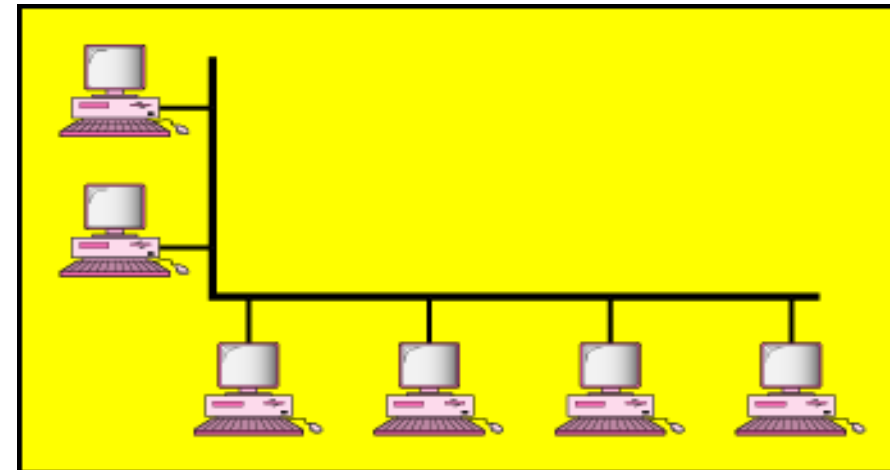
“A LAN connects devices within a limited geographical area such as a room, building, office, or campus.”

Characteristics

- Coverage: Building/Campus
- High data rate
- Privately owned
- Low latency

Examples

- Home Wi-Fi network
- Computer laboratory
- Office network



a. Single-building LAN

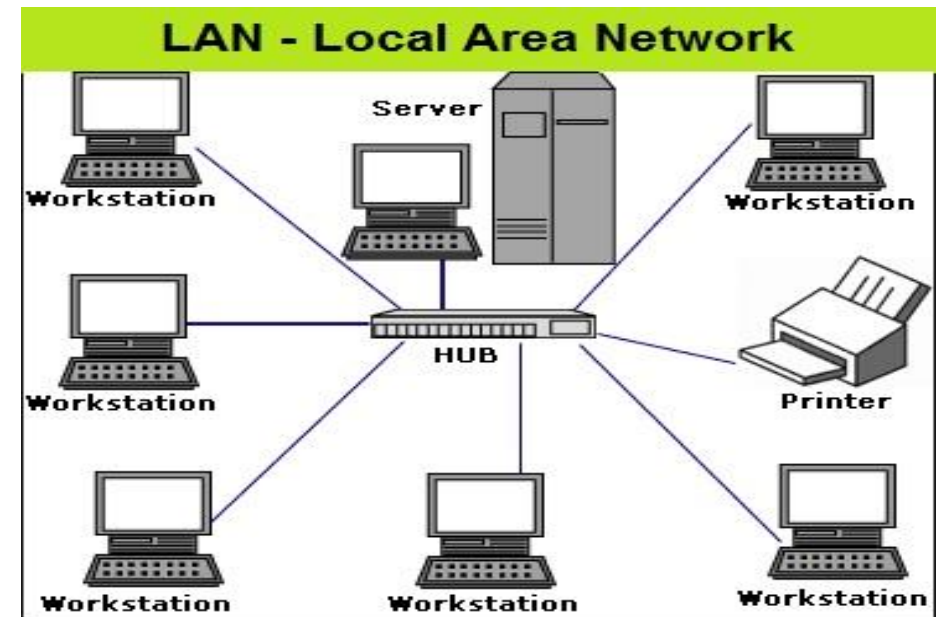


LOCAL AREA NETWORK (LAN)

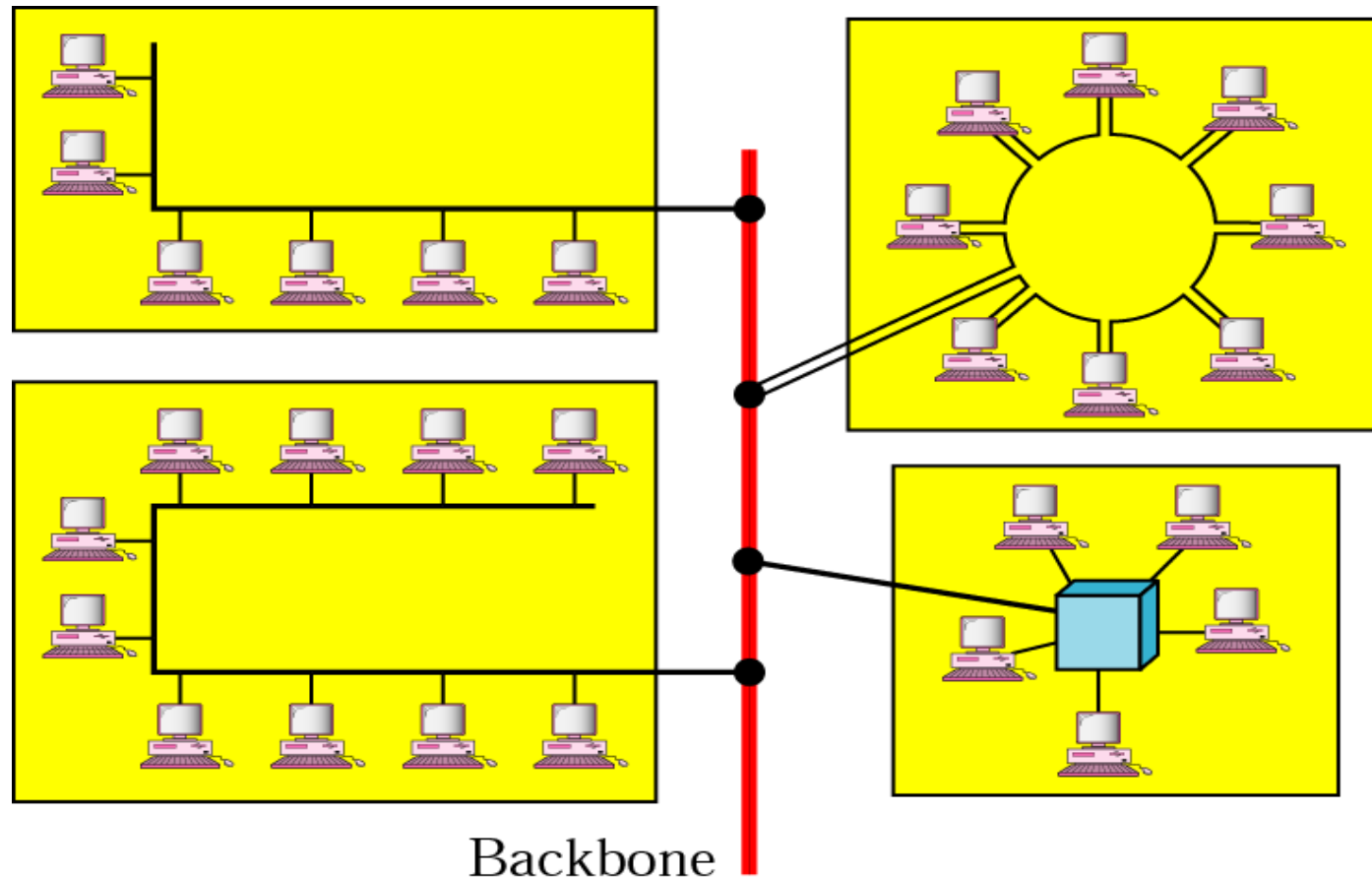
- A LAN is a network that is used for communicating among computer devices, usually within an office building or home.
- LAN's enable the sharing of resources such as files or hardware devices that may be needed by multiple users.
- LANs can be built with relatively inexpensive hardware, such as hubs, network adapters and Ethernet cables.
- Is limited in size, typically spanning a few hundred meters, and no more than a mile.
- Is fast, with speeds from 10 Mbps to 10Gbps
- Has lower cost compared to MAN's or WAN's

LOCAL AREA NETWORK (LAN) Cont...

- A Communication medium used for LAN has twisted pair cables and coaxial cables. It covers a short distance, and so the error and noise are minimized.
- Nodes in a LAN are linked together with a certain topology. These topologies include:
Bus, Ring, Star.
- LAN's can be either wired or wireless.



LOCAL AREA NETWORK (LAN)



b. Multiple-building LAN

METROPOLITAN AREA NETWORK (MAN)

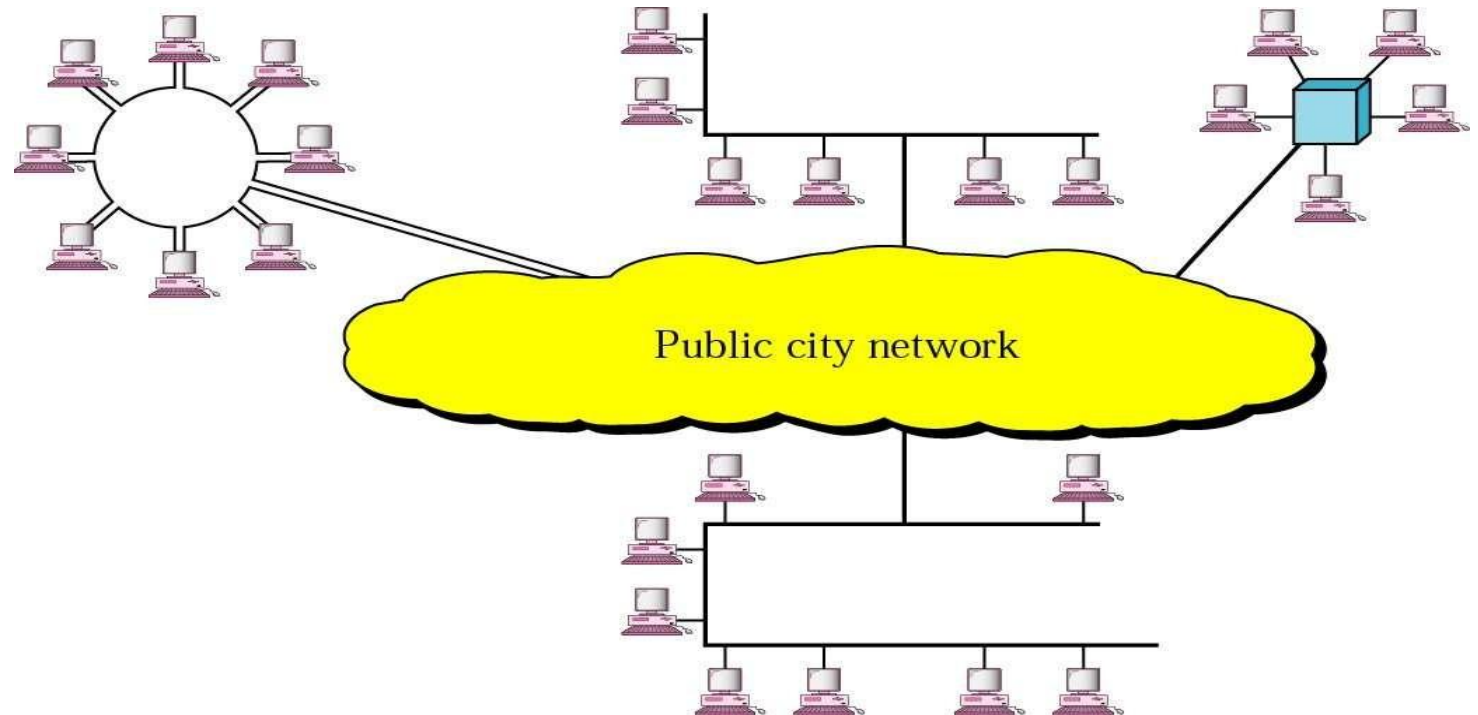
“A MAN connects multiple LANs across a city or metropolitan area.”

Characteristics

- Coverage: City
- High-speed backbone
- Often managed by service providers

Examples

- City-wide cable TV network
- Municipal networks
- Metropolitan broadband services



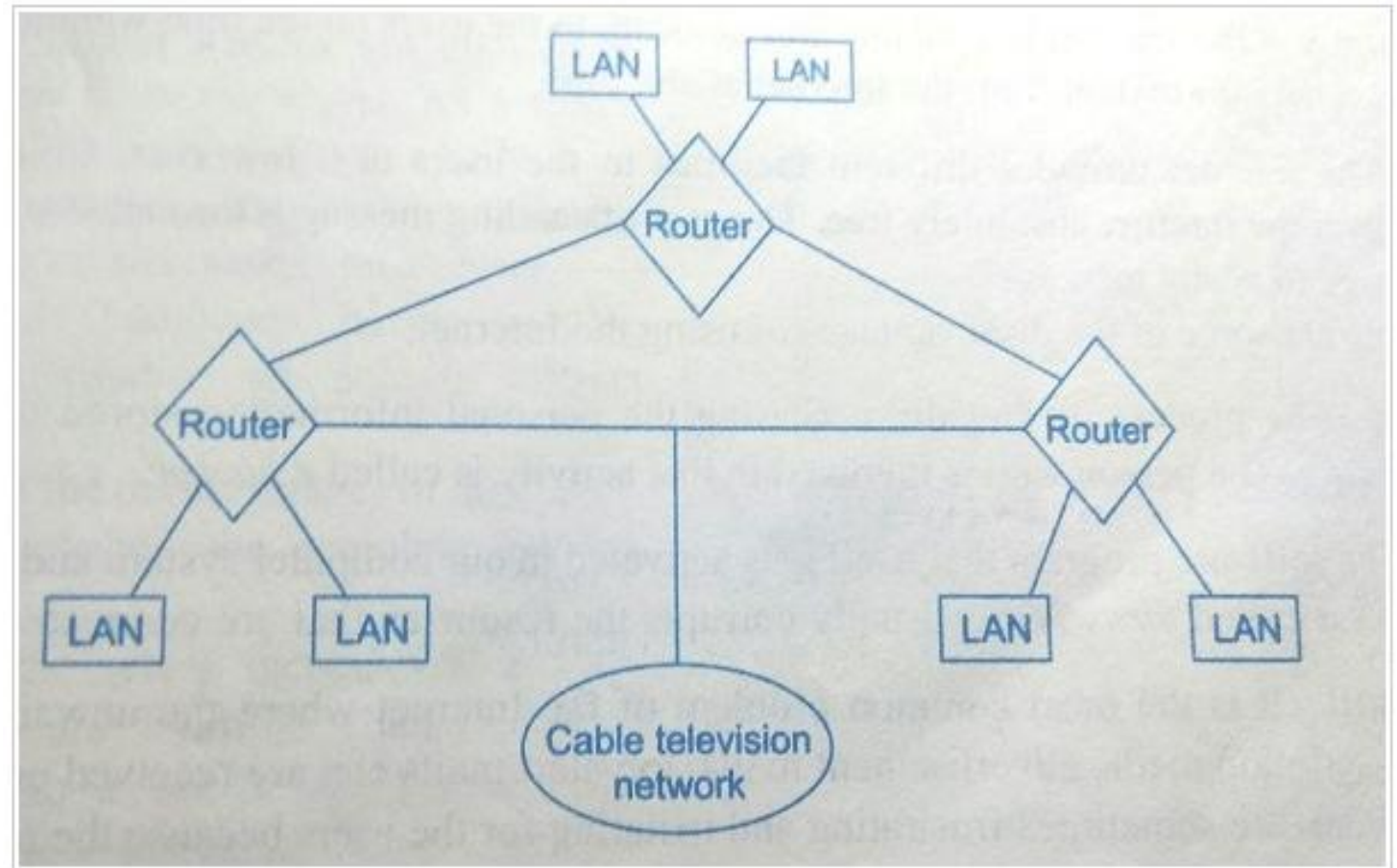


METROPOLITAN AREA NETWORK (MAN)

- A data network designed for a town or city.
- In terms of geographic breadth, MANs are larger than local-area networks (LANs), but smaller than wide-area networks (WANs).
- MANs are usually owned by large organizations to interconnect its various branches across a city.
- MAN is generally connected with the help of optical fibers, copper wires etc.
- One of the most common examples of MAN is **cable television** network within a city

METROPOLITAN AREA NETWORK (MAN)

- A network device known as **router** is used to connect the LANs together. The router directs the information packets to the desired destination.





WIDE AREA NETWORK (WAN)

“A WAN connects networks over large geographical areas such as countries or continents.”

Characteristics

- Coverage: Country/World
- Uses public and private communication links
- Higher latency than LAN

Examples

- Internet
- Banking networks
- Airline reservation systems

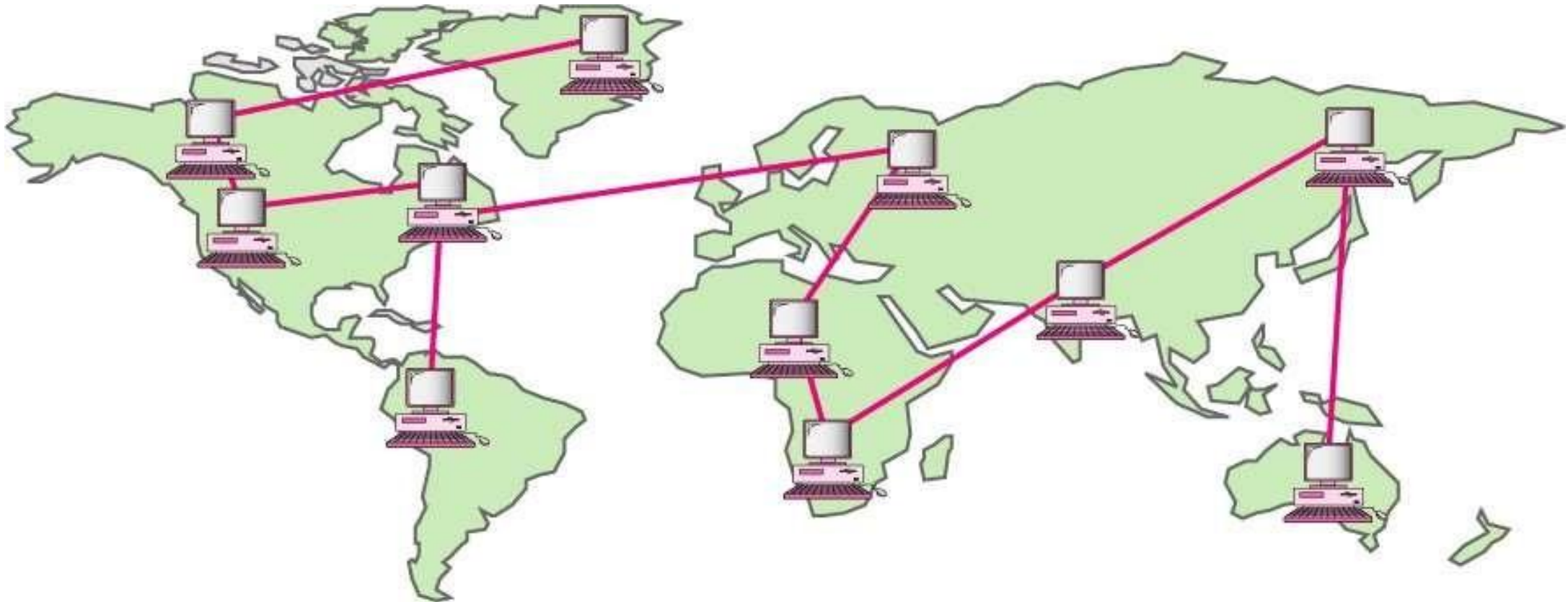


WIDE AREA NETWORK (WAN)

- A computer network that spans a relatively large geographical area **such as continent, country, etc.**
- Typically, a WAN consists of two or more local-area networks (LANs).
- Computers connected to a wide-area network are often connected through public networks, such as the **telephone** system.
- They can also be connected through leased lines or **satellites**.
The largest WAN in existence is the **Internet**.

WIDE AREA NETWORK (WAN)

“The Internet is the world's largest WAN, connecting millions of networks globally using TCP/IP.”





NETWORK PROTOCOLS

- Imagine two people speaking different languages:
 - One speaks English.
 - One speaks Hindi.
- Communication becomes difficult.
- Similarly, computers also need a common set of rules to communicate.
- Communication is successful only when all parties follow the same rules.

Thus, ***“A protocol is a set of rules governing communication between devices.”***



NETWORK PROTOCOLS

Need of Network Protocols?

- Define message format.
- Control data transmission.
- Handle errors.
- Ensure reliable communication.

Examples

- Human communication rules
- Traffic rules
- Computer network rules



ELEMENTS OF A PROTOCOL

Every protocol consists of three key elements:

1. **Syntax:** Structure and format of data.
2. **Semantics:** Meaning of each field.
3. **Timing:** When and how fast data should be transmitted.

Protocol	Purpose
HTTP	Web communication
HTTPS	Secure web communication
FTP	File transfer
SMTP	Email sending
POP3/IMAP	Email retrieval
DNS	Domain name resolution
DHCP	Automatic IP assignment



NETWORK MODELS

Problem Statement:

- Imagine two computers communicating over a network. Several tasks must be performed:
 - Data generation
 - Data formatting
 - Error detection
 - Routing
 - Addressing
 - Data transmission
- If all these tasks are handled by a single program, the system becomes:
 - Complex
 - Difficult to maintain
 - Difficult to troubleshoot

Solution:

Divide communication into smaller manageable tasks called **layers**.

LAYERED APPROACH

In the study of computer networks, it is essential to study the way our networks work. Computer networks are operated by network models; most prominently the OSI-RM and the TCP/ IP Model.

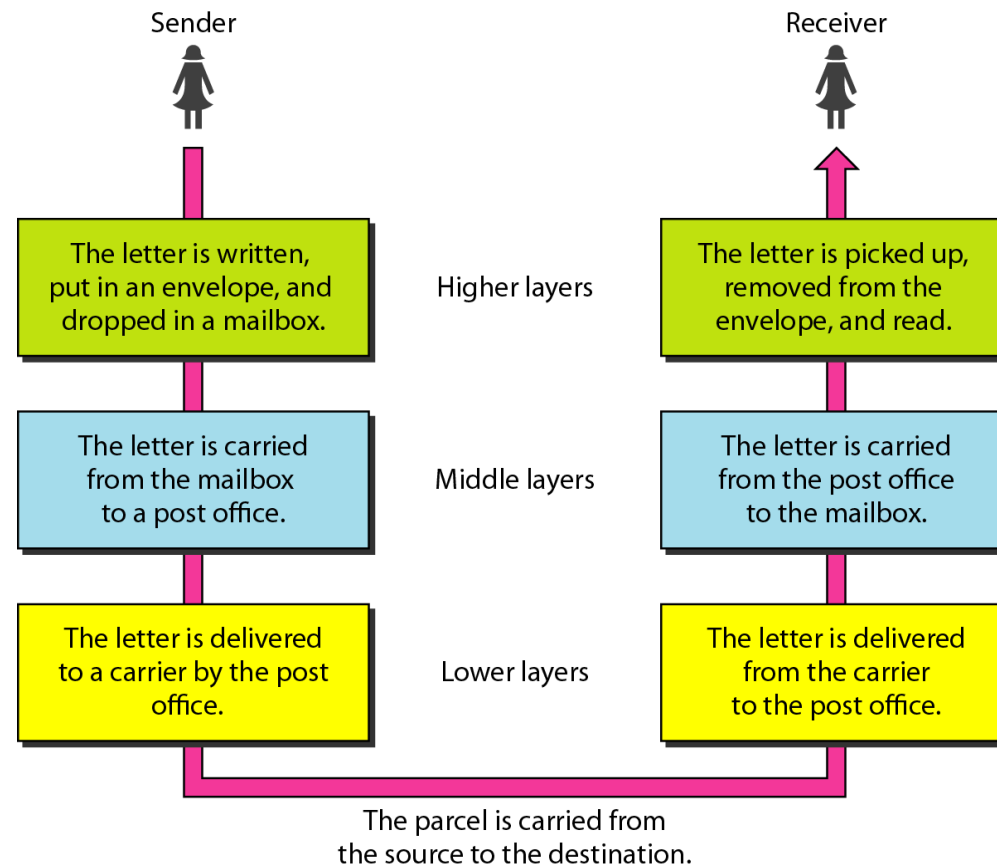
Concept of Layered Task:

- i. The main objective of a computer network is to be able to transfer the data from sender to receiver. This task can be done by breaking it into small sub tasks, each of which are well defined.
- ii. Each subtask will have its own process or processes to do and will take specific inputs and give specific outputs to the subtask before or after it. In more technical terms we can call these sub tasks as layers.
- iii. In general, every task or job can be done by dividing it into sub task or layers.

Consider the example of sending a letter where the sender is in City A and receiver is in city B

LAYERED APPROACH

*We use the concept of **layers** in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.*



The figure shows:

- a. Sender, Receiver & Carrier
- b. Hierarchy of layers



LAYERED APPROACH

Layered architecture is a design approach in which network communication is divided into multiple layers, each performing a specific function.

Key Concepts

- Each layer performs a specific task.
- Each layer provides services to the layer above.
- Each layer receives services from the layer below.
- Communication occurs between corresponding layers on different systems.

Benefits

- Modularity
- Scalability
- Standardization



THE OSI MODEL

Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

**ISO is the organization.
OSI is the model.**

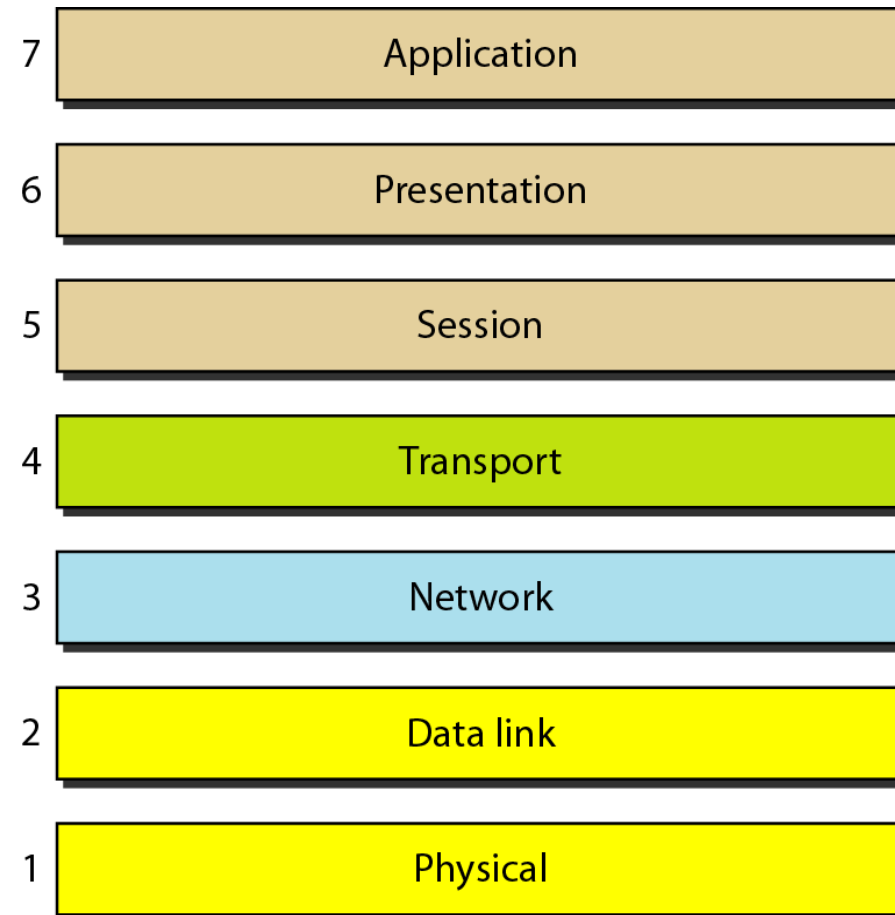
Purpose: Provides a standard framework for communication between different computer systems.

Characteristics

- Seven-layer model
- Conceptual reference model
- Ensures interoperability



SEVEN LAYERS OSI MODEL





OSI MODEL

- It is a network model that defines the protocols for network communications. It is a hierarchical model that groups its processes into layers. It has 7 layers as follows: (Top to Bottom)

- 1. Application Layer**
- 2. Presentation Layer**
- 3. Session Layer**
- 4. Transport Layer**
- 5. Network Layer**
- 6. Data Link Layer**
- 7. Physical Layer**

- Each layer has specific duties to perform and must co-operate with the layers above and below it.



Layered Architecture of OSI MODEL

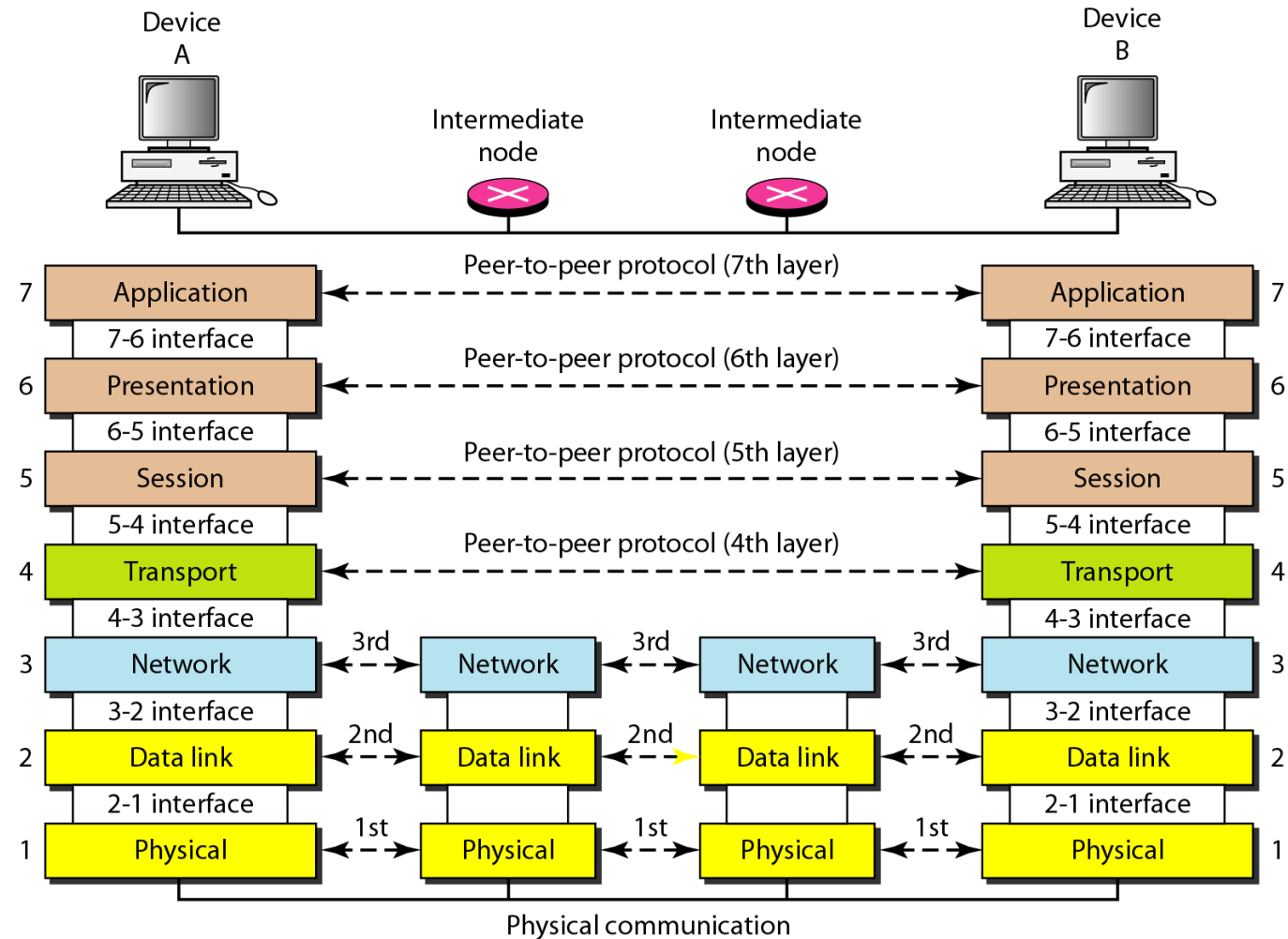
- The OSI model has 7 layers each with its own dedicated task.
- A message sent from Device A to Device B passes must pass through all layers at A from top to bottom then all layers at B from bottom to top.
- At Device A, the message is sent from the top layer i.e., Application Layer, then all the layers till it reaches its physical layer and then it is transmitted through the transmission medium.
- At Device B, the message received by the physical layer passes through all its other layers and moves upwards till it reaches its Application Layer.
- As the message travels from device A to device B, it may pass through many intermediate nodes. These intermediate nodes usually involve only the first three layers of the OSI model.



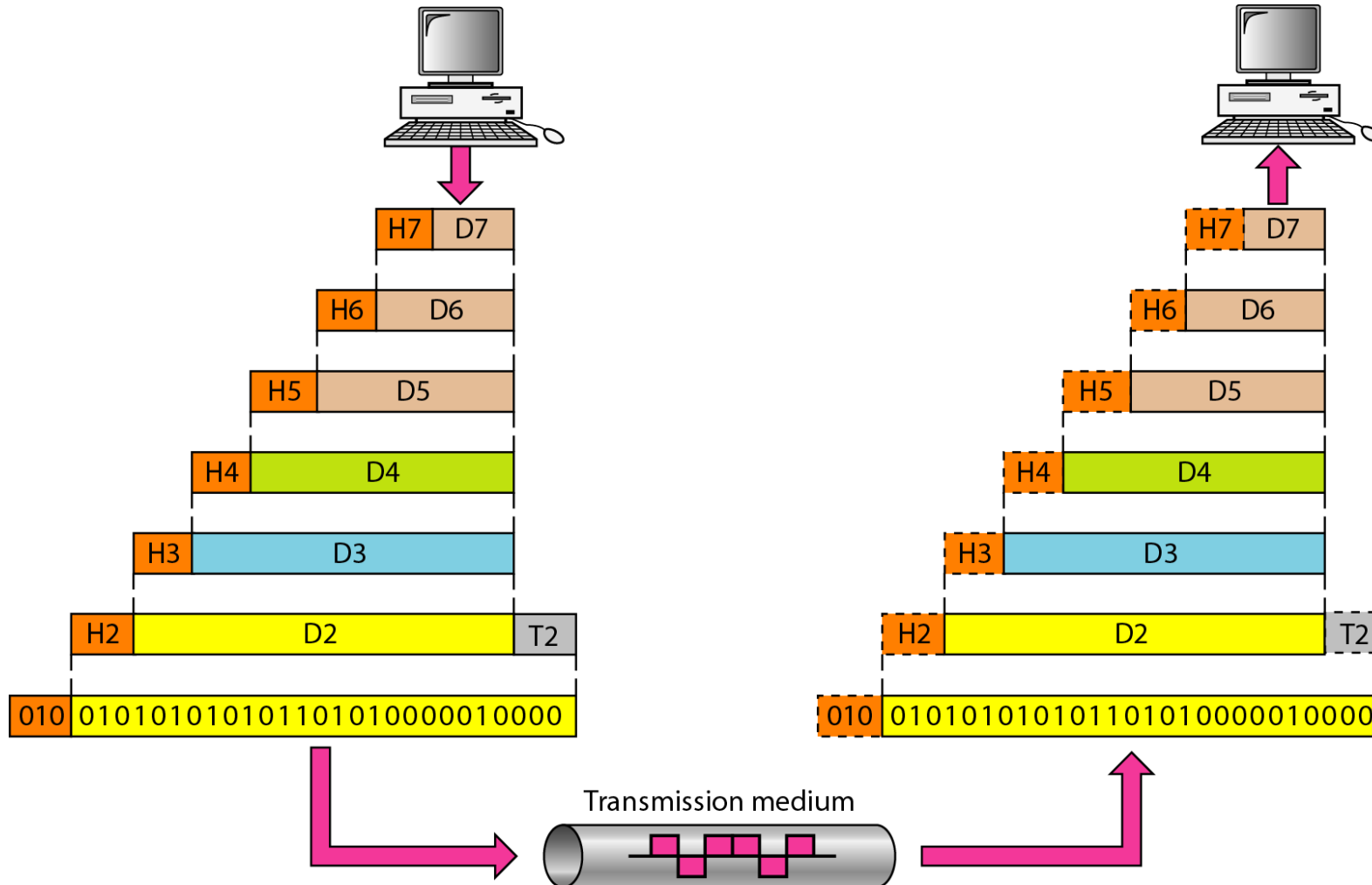
Communication & Interfaces:

- For communication to occur, each layer in the sending device adds its own information to the message it receives from the layer just above it and passes the whole package to the layer just below it. Each layer in the receiving device removes the information added at the corresponding layer and sends the obtained data to the layer above it.
- Every Layer has its own dedicated function or services and is different from the function of the other layers.
- On every sending device, each layer calls upon the service offered by the layer below it.
- On every receiving device, each layer calls upon the service offered by the layer above it.
- Between two devices, the layers at corresponding levels communicate with each other. i.e layer 2 at receiving end can communicate and understand data from layer 2 of sending end. This is called peer –to – peer communication.
- For this communication to be possible between every two adjacent layers there is an interface. An interface defines the service that a layer must provide. Every layer has an interface to the layer above and below it.

INTERACTION BETWEEN LAYERS OF OSI MODEL



EXCHANGE OF INFORMATION USING OSI MODEL



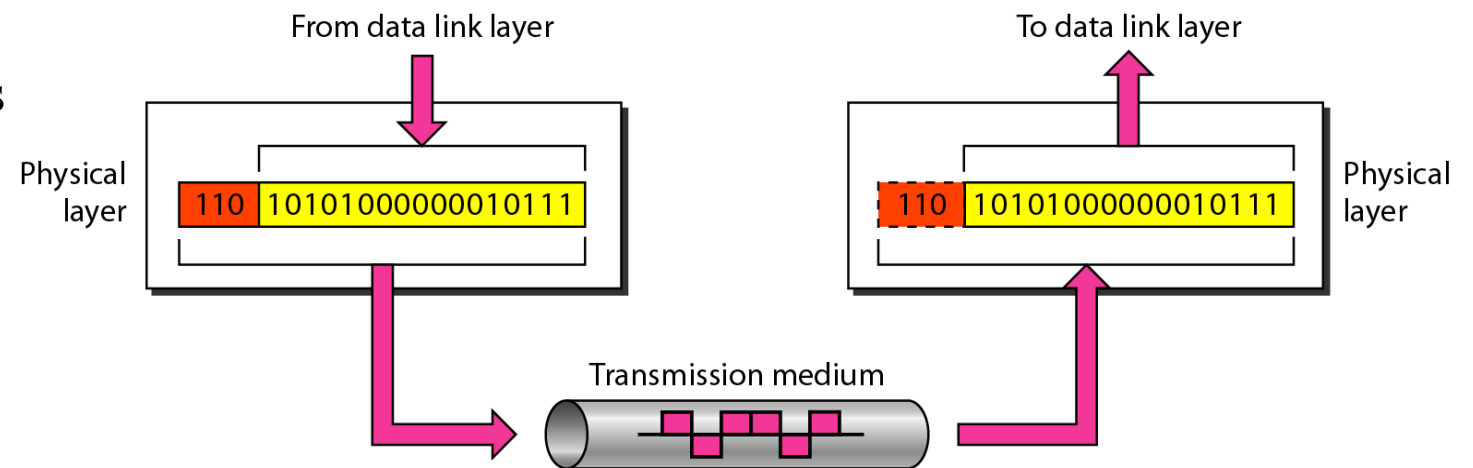
PHYSICAL LAYER

Functions

- Transmission of raw bits
- Electrical and mechanical specifications
- Data rate and synchronization

Examples

- Ethernet cables
- Optical fiber
- Wireless signals



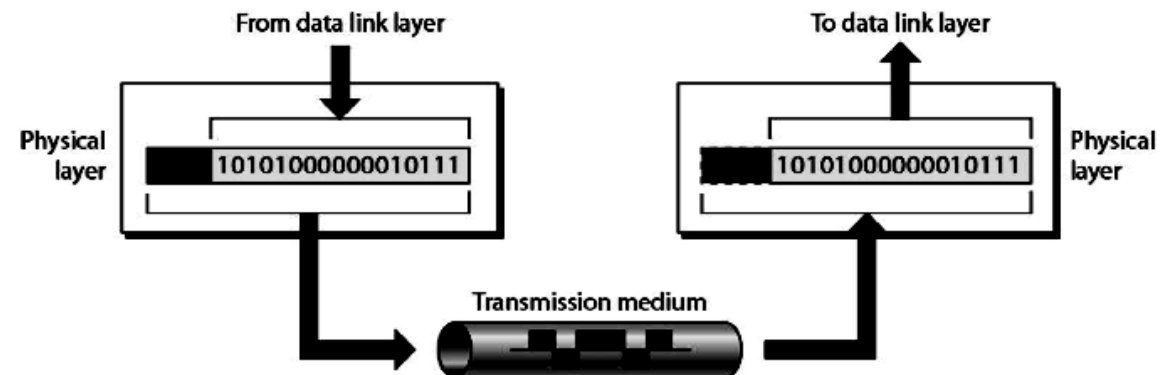
The physical layer is responsible for movements of individual bits from one hop (node) to the next.

PHYSICAL LAYER

- I. The Physical Layer provides a standardized interface to physical transmission media, including :
 - a. Mechanical specification of electrical connectors and cables, e.g., maximum cable length
 - b. Electrical specification of transmission line
 - c. Bit-by-bit or symbol-by-symbol delivery

- II. On the sender side, the physical layer receives the data from Data Link Layer and encodes it into signals to be transmitted onto the medium. On the receiver side, the physical layer receives the signals from the transmission medium, decodes it back into data and sends it to the Data Link Layer.

Main Responsibility:
Bit by bit transmission.



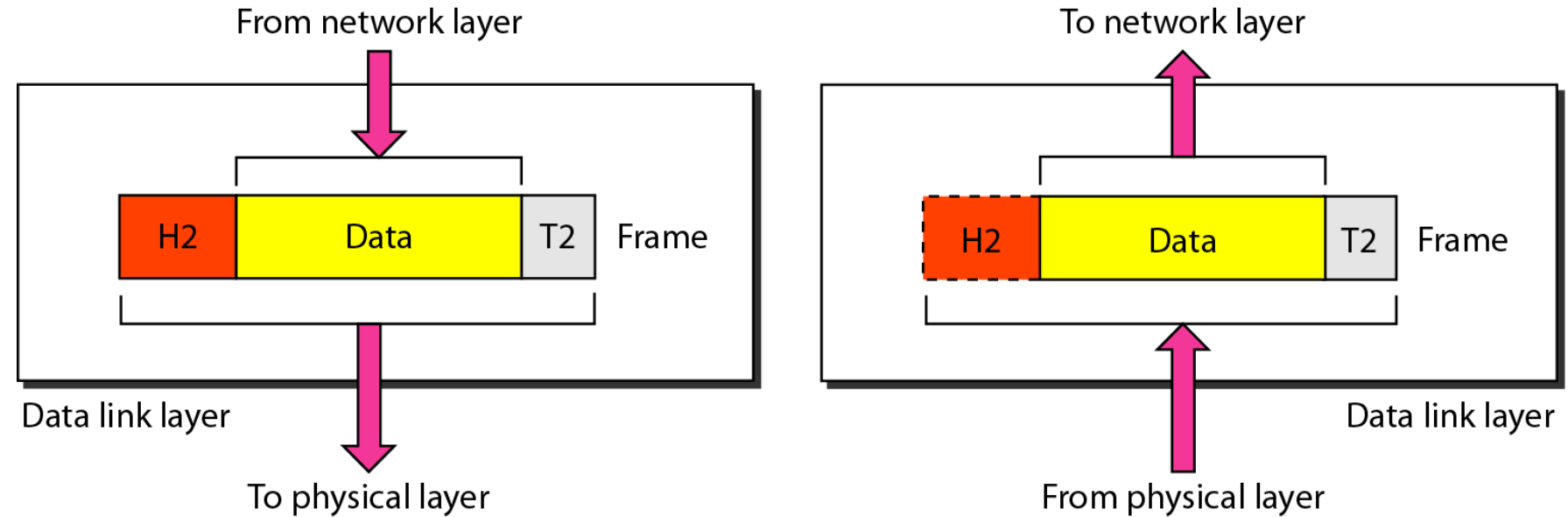
DATA LINK LAYER

Functions

- Framing Error detection
- Flow control
- MAC addressing

Examples

- Ethernet Frame



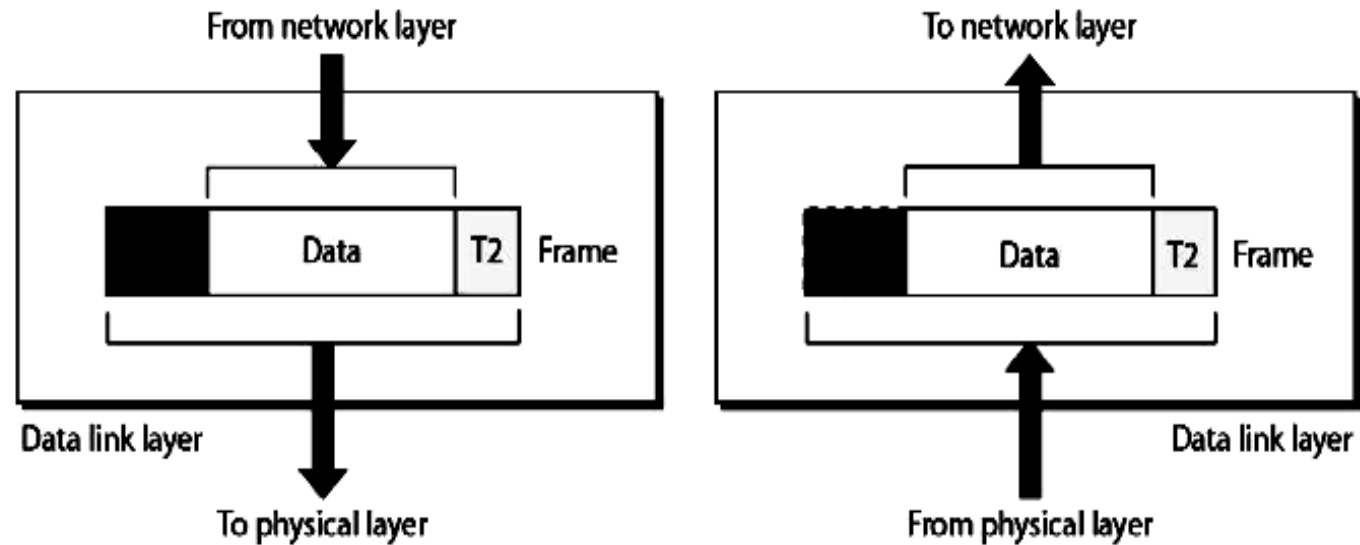
The data link layer is responsible for moving frames from one hop (node) to the next.

Data Link Layer

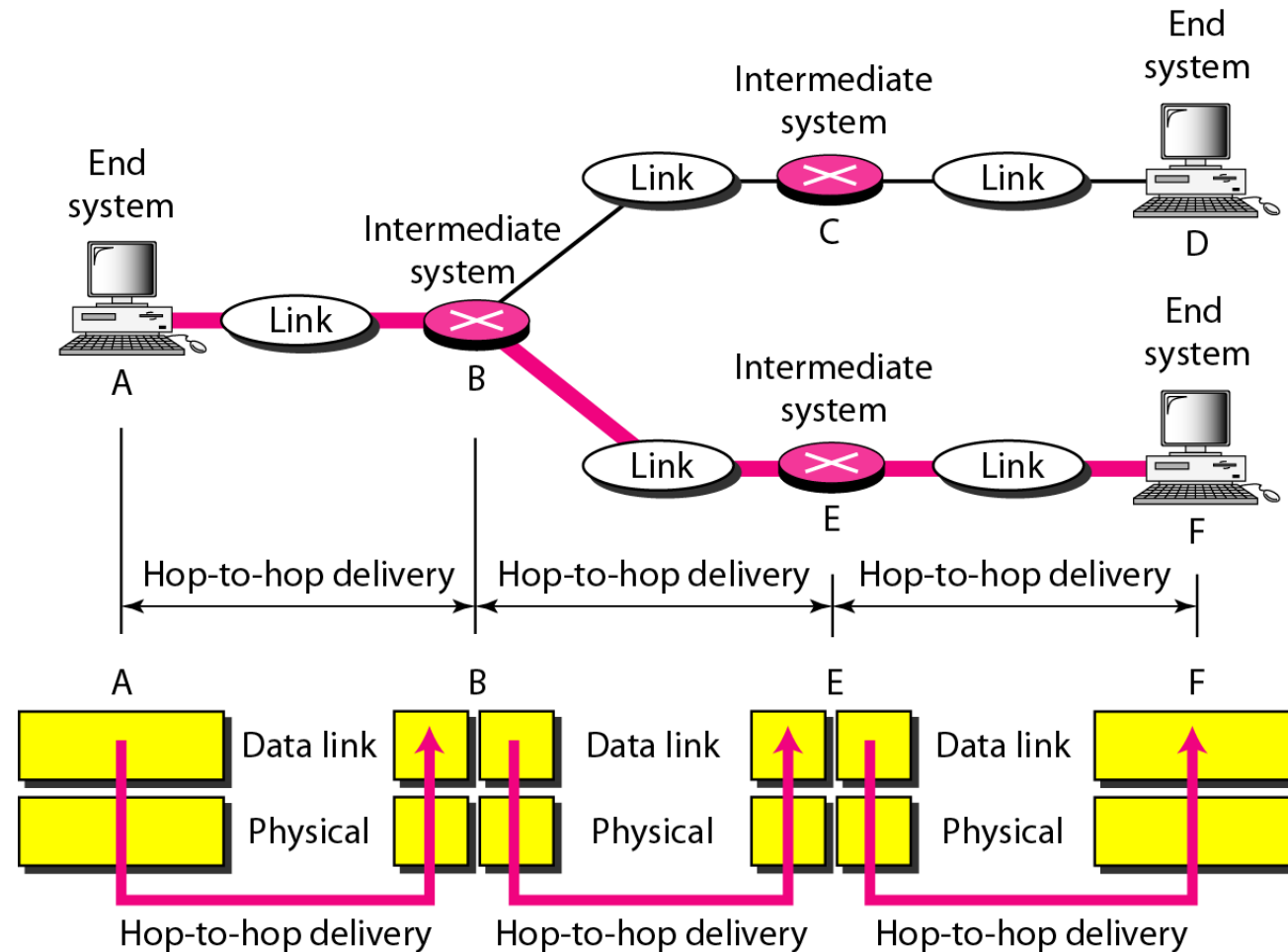
I. The Data Link layer adds reliability to the physical layer by providing error detection and correction mechanisms.

II. On the **sender** side, the Data Link layer receives the data from Network Layer and divides the **stream of bits into fixed size** manageable units called as **Frames** and sends it to the physical layer. On the **receiver** side, the data link layer receives the stream of bits from the physical layer and regroups them into frames and sends them to the Network layer. This process is called **Framing**. It is shown in the figure below:

Main Responsibility:
transmission of frames.



DATA LINK LAYER: HOP TO HOP DELIVERY



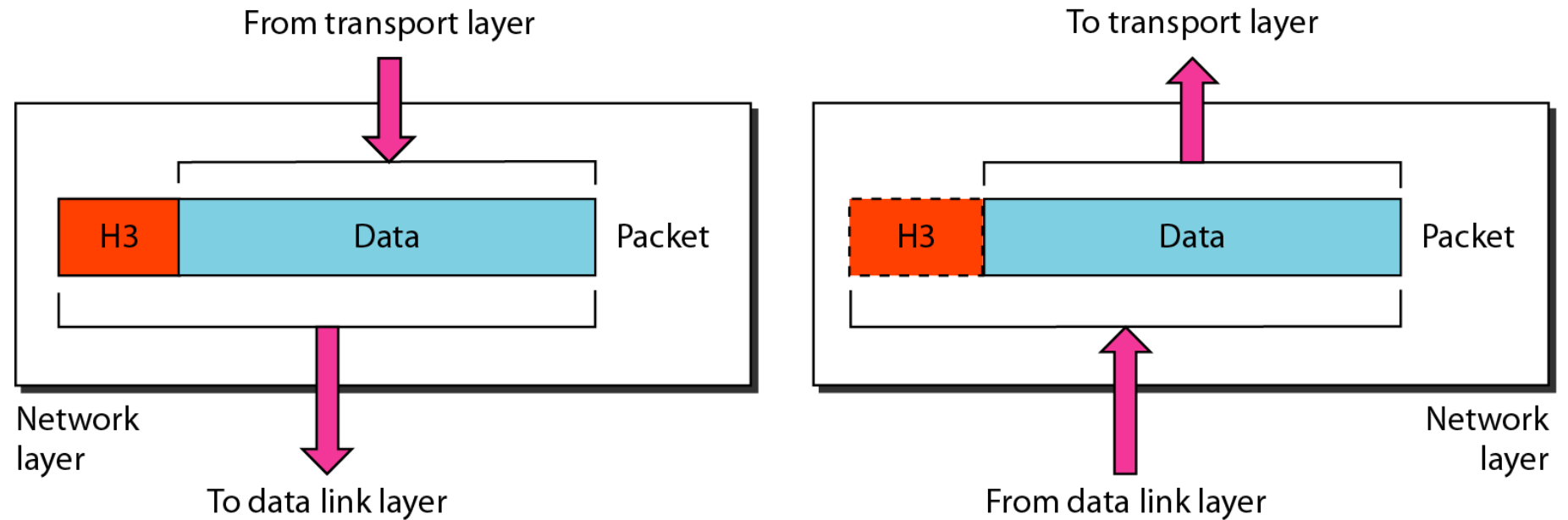
NETWORK LAYER

Functions

- Logical addressing
- Routing
- Path selection

Examples

- IP
- Router



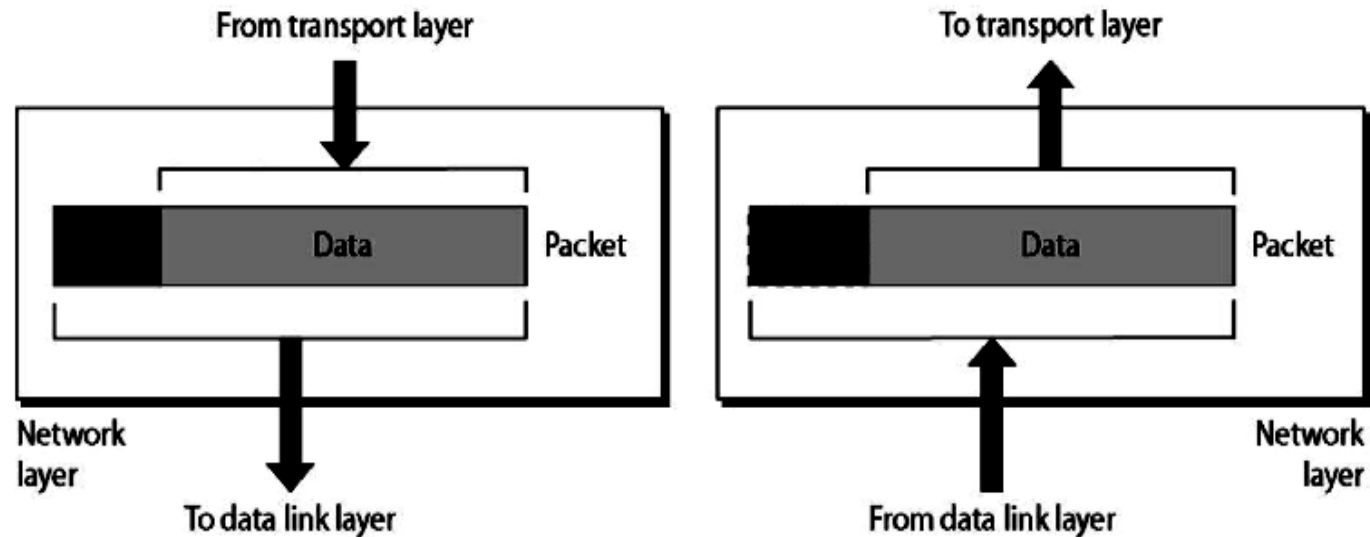
The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Network Layer

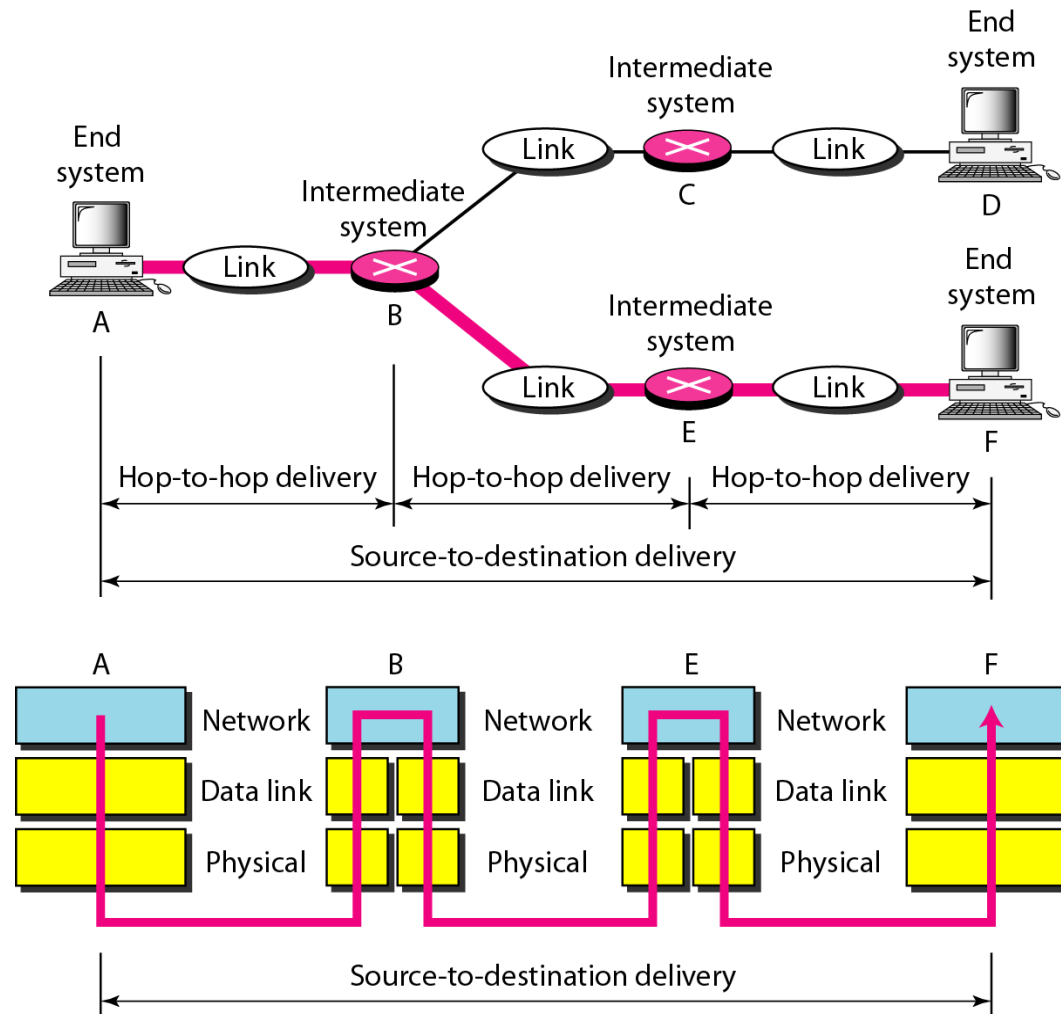
I. The network layer makes sure that the data is delivered to the receiver despite multiple intermediate devices.

II. The network layer at the **sending** side accepts data from the transport layer, **divides it into packets, adds addressing information in the header** and passes it to the data link layer. At the **receiving** end, the network layer **receives the frames** sent by data link layer, **converts them back into packets, verifies the physical address** (verifies if the receiver address matches with its own address) and then send the packets to the transport layer.

Main Responsibility:
transmission of packets from
source to destination



NETWORK LAYER: SOURCE TO DESTINATION DELIVERY



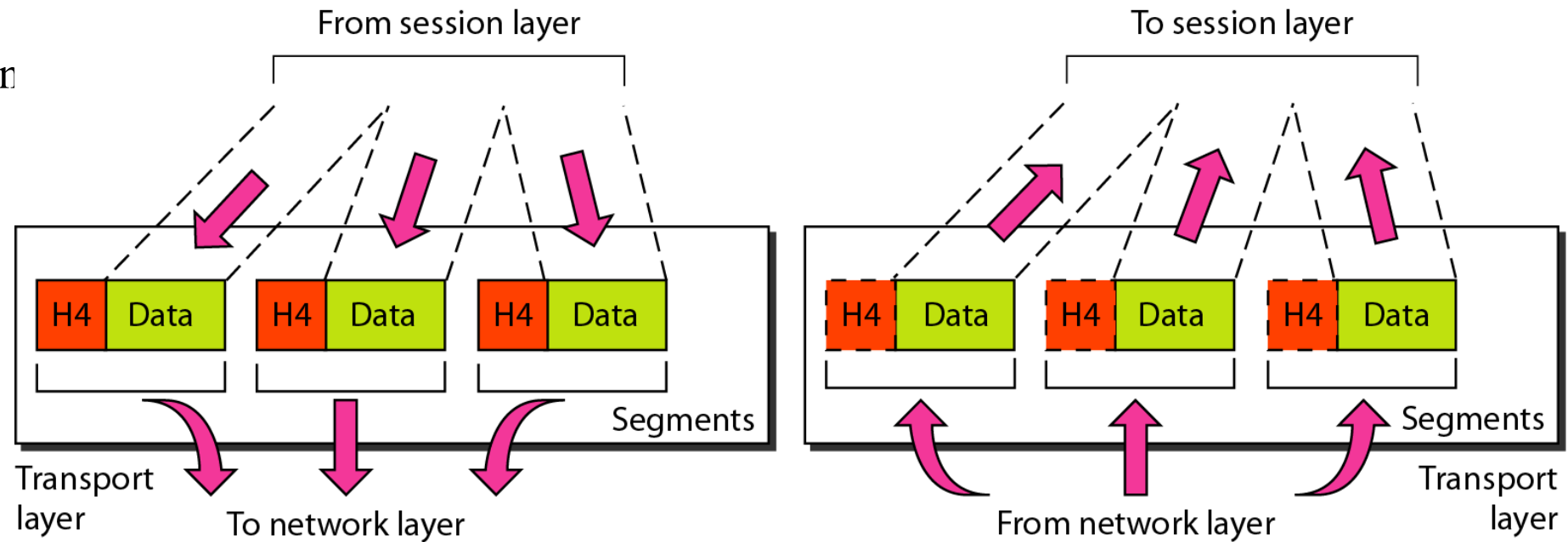
TRANSPORT LAYER

Functions

- End-to-end communication
- Error recovery
- Segmentation
- Flow control

Examples

- TCP
- UDP



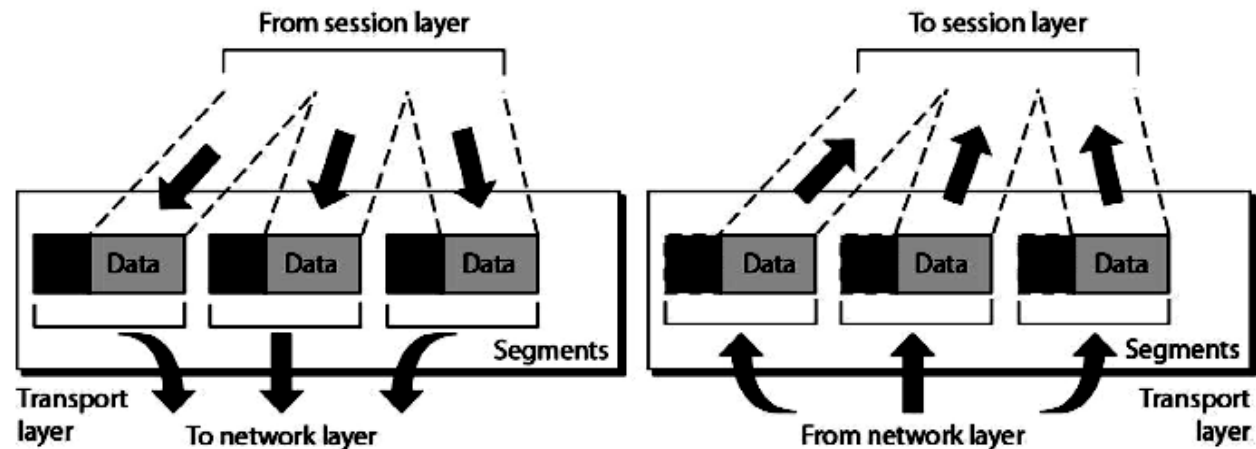
The transport layer is responsible for the delivery of a message from one process to another.

Transport Layer

- I. A logical address at network layer facilitates the transmission of data from source to destination device. But the **source** and the **destination** both may be having multiple processes communicating with each other. Hence, it is important to deliver the data not only from the sender to the receiver but from the **correct process on the sender to the correct process on the receiver**. The transport layer takes care of **process-to-process delivery** of data and makes sure that it is intact and in order.
- II. At the **sending** side, the transport layer receives data from the session layer, divides it into units called **segments** and sends it to the network layer. At the **receiving** side, the transport layer receives packets from the network layer, converts and arranges into proper sequence of **segments** and sends it to the session layer.

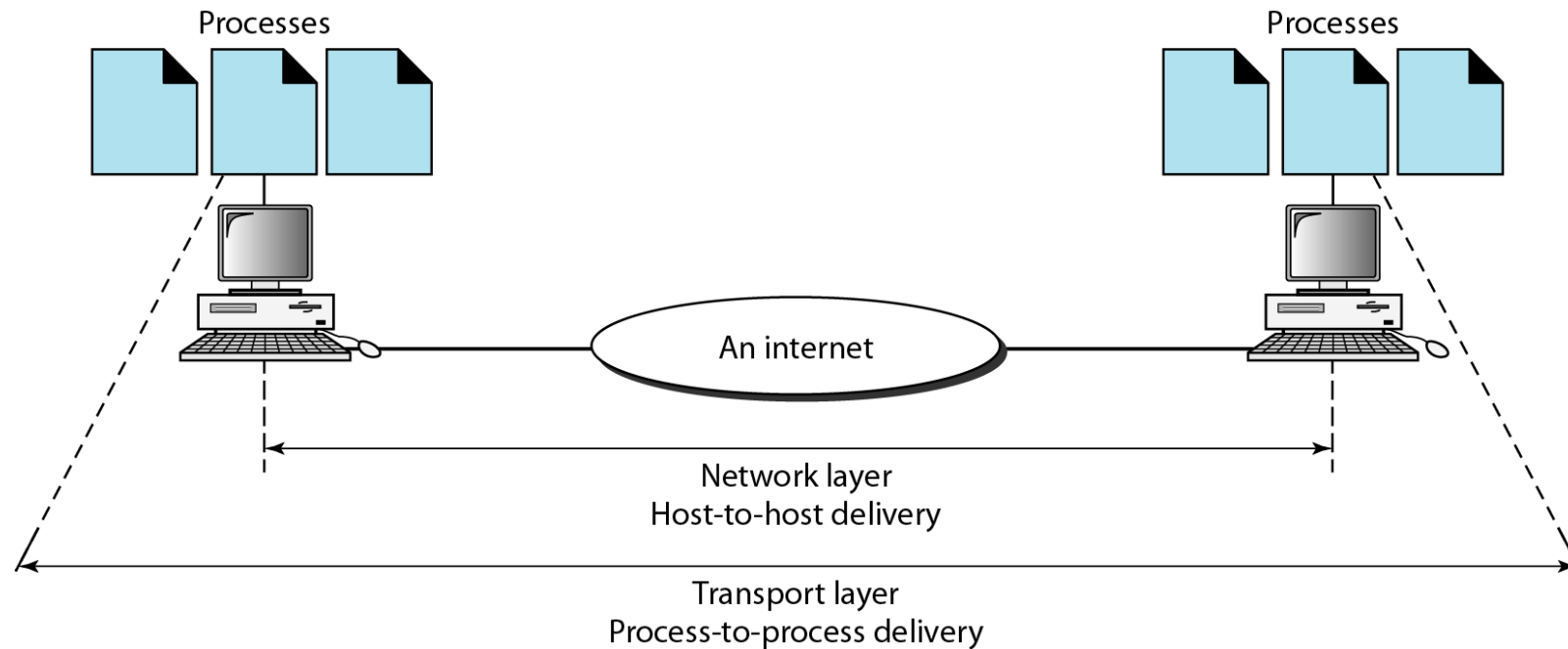
Main Responsibility

process to process delivery of the entire message. (segments)





TRANSPORT LAYER: PROCESS TO PROCESS DELIVERY



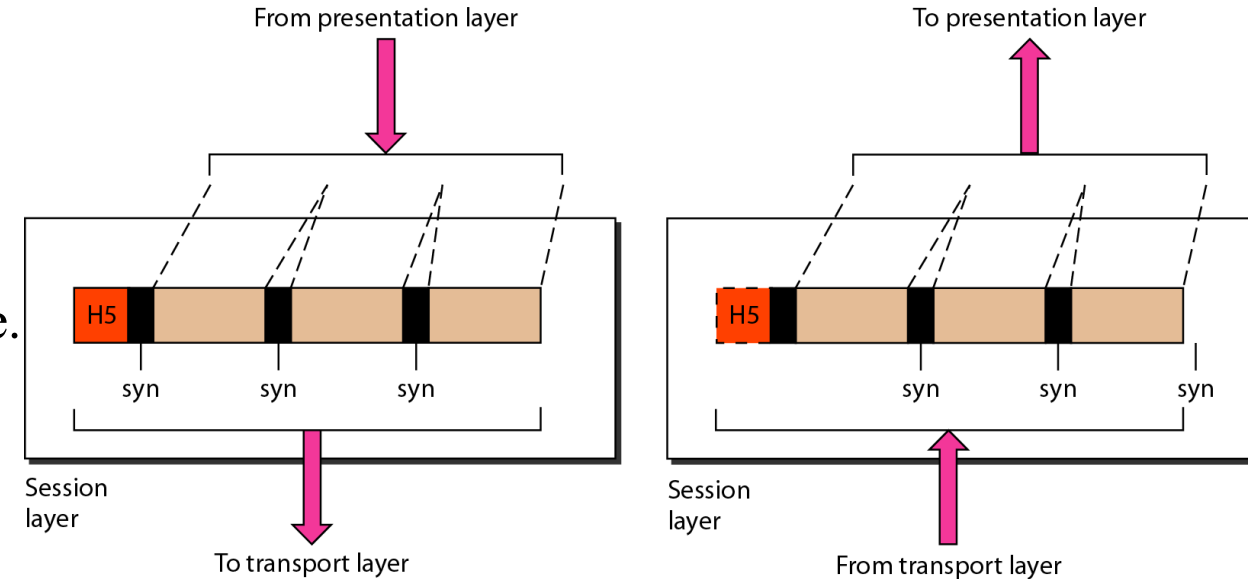
SESSION LAYER

Functions

- Establishes communication sessions.
- Maintains and synchronizes data exchange.
- Controls dialog between communicating devices.
- Terminates sessions after communication is complete.

Examples

- Remote login sessions.
- Video conferencing sessions.
- Online banking transactions.



The session layer is responsible for dialog control and synchronization.

Session Layer

I. The session layer establishes a session between the communicating devices called **dialog and synchronizes their interaction**. It is the responsibility of the session layer to establish and synchronize the dialogs. It is also called the network dialog controller.

II. The session layer at the **sending** side accepts data from the presentation layer **adds checkpoints** to it called **sync bits** and passes the data to the transport layer. At the **receiving** end, the session layer receives data from the transport layer **removes the checkpoints** inserted previously and passes the data to the presentation layer.

III. The checkpoints or synchronization points is a way of informing the status of the data transfer.

Example: A checkpoint after first 500 bits of data will ensure that those 500 bits are not sent again.

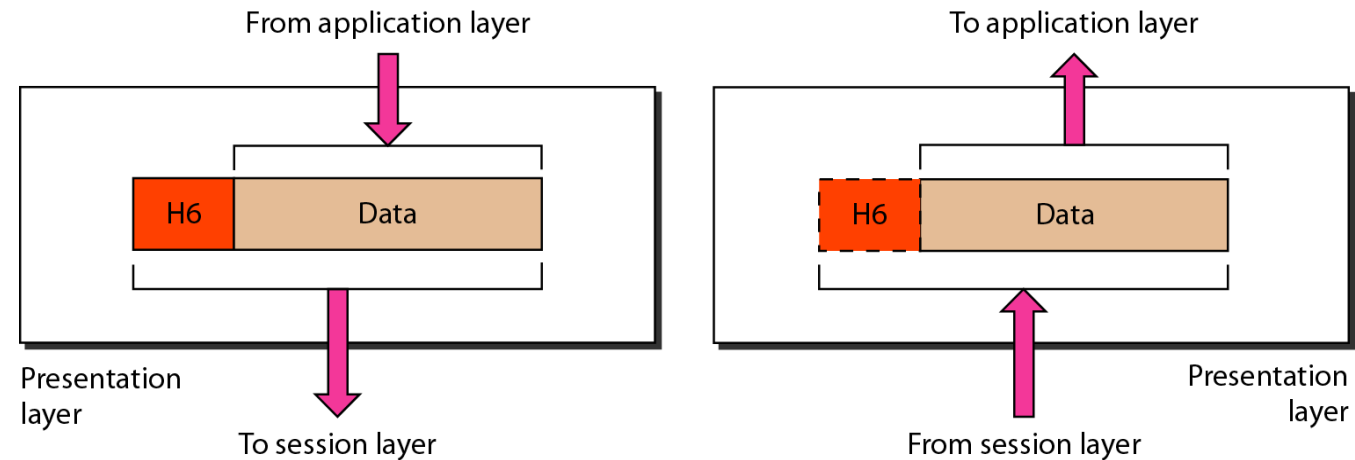
PRESENTATION LAYER

Functions

- Data translation and format conversion.
- Data encryption and decryption.
- Data compression and decompression.
- Ensures compatibility between different systems.

Examples

- JPEG, PNG image formats.
- MP3 audio format.
- SSL/TLS encryption used in HTTPS.

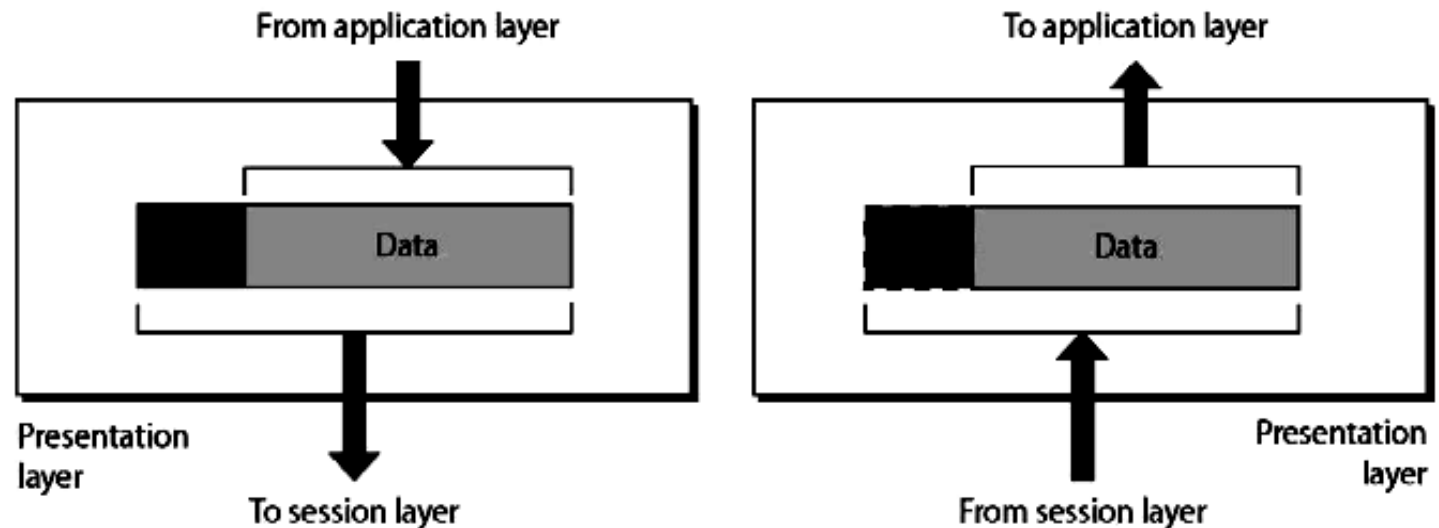


The Presentation Layer ensures that data is presented in a format that the receiving application can understand.

Presentation Layer

- I. The communicating devices may be having different platforms. The presentation layer performs **translation, encryption and compression of data**.
- II. The presentation layer at **sending** side receives the data from the application layer **adds header** which contains information related to encryption and compression and sends it to the session layer. At the **receiving** side, the presentation layer receives data from the session layer **decompresses** and **decrypts** the data as required and translates it back as per the encoding scheme used at the receiver.

Main responsibility:
translation, compression
and encryption.



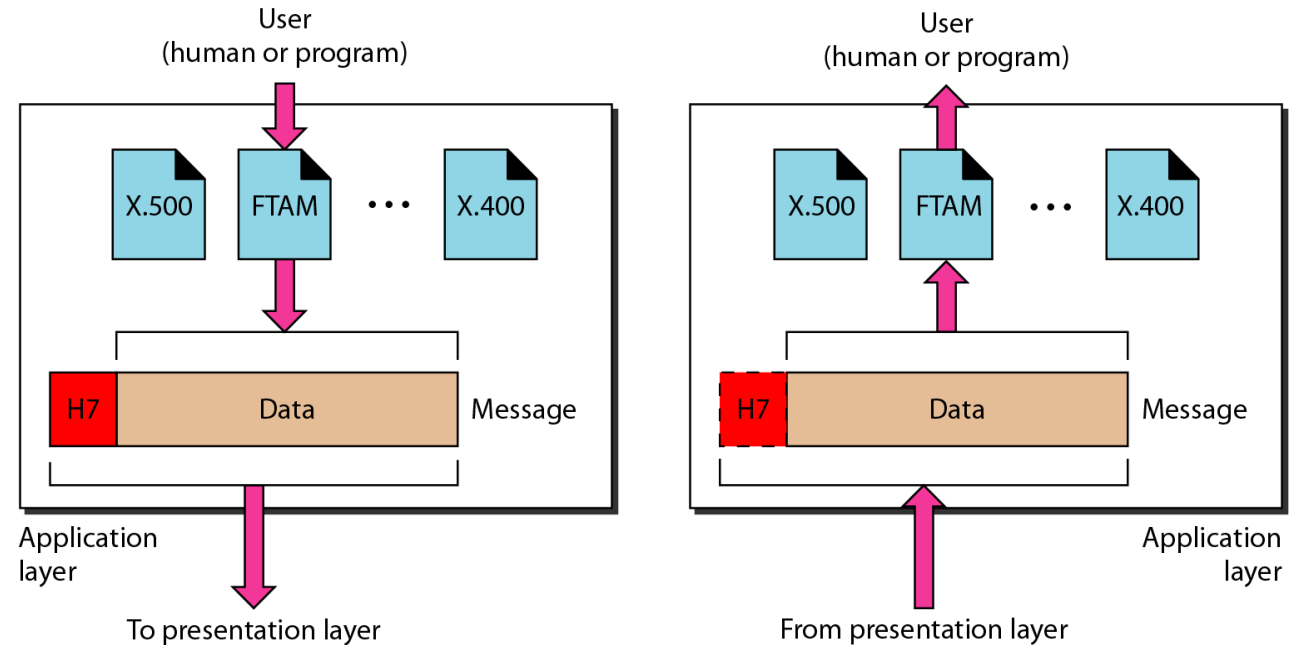
APPLICATION LAYER

Functions

- Provides user access to network services.
- Supports file transfer, email, and web browsing.
- Interface between applications and the network.

Examples

- Web browsers.
- Email applications.
- Online file sharing services.

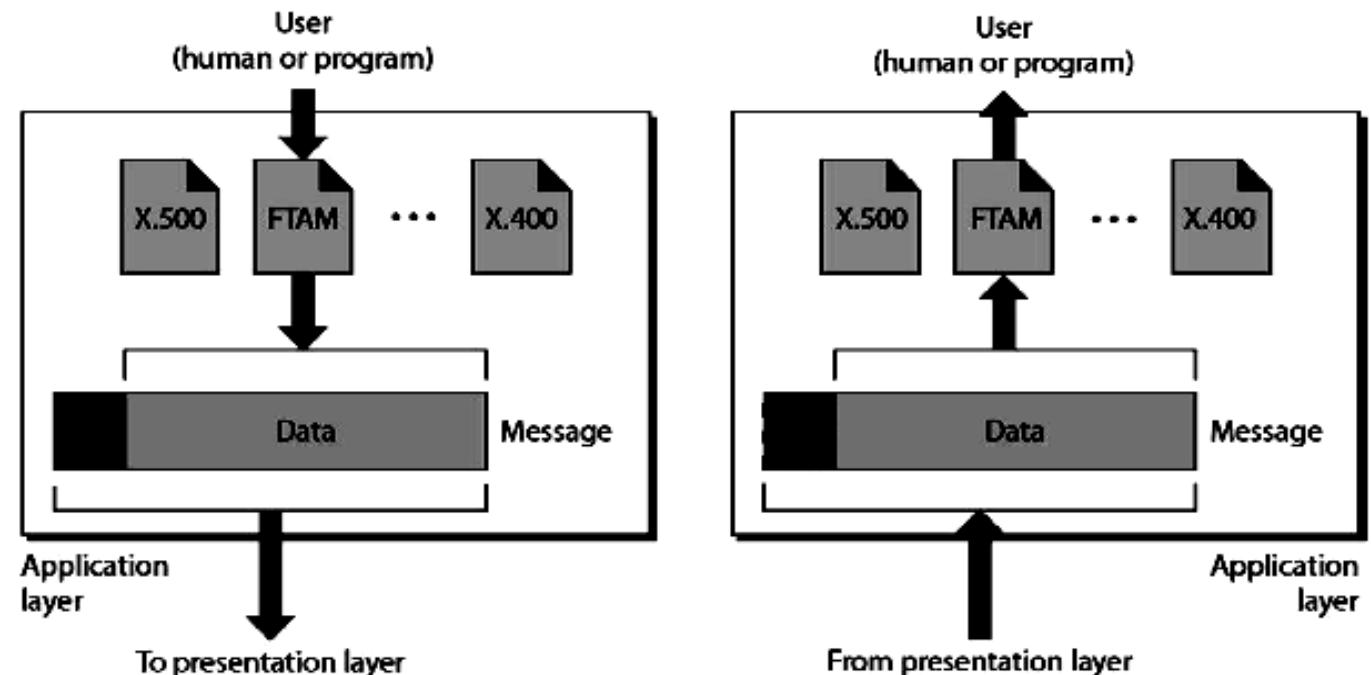


The application layer is responsible for providing services to the user and access to network resources.

Application Layer

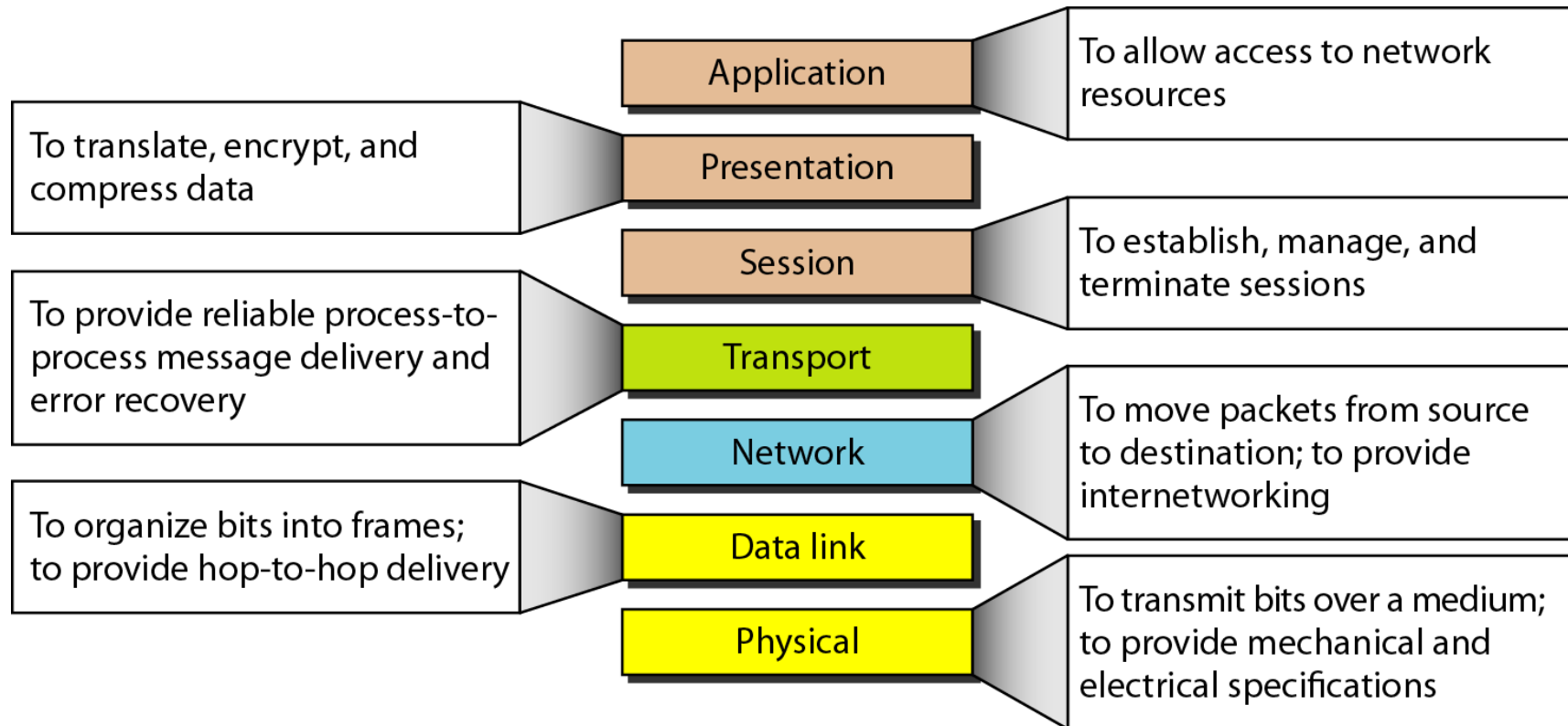
- I. The application layer enables the user to communicate its data to the receiver by providing certain **services**. It provides user services like user login, naming network devices, formatting messages, and e-mails, transfer of files etc.
- II. FTAM (File transfer, access and management) provides access to files stored on remote computers and mechanism for transfer and manage them locally.

Main Responsibility:
provide access to
network resources.





SUMMARY OF LAYERS





ADVANTAGES AND LIMITATIONS OF OSI MODEL

Advantages

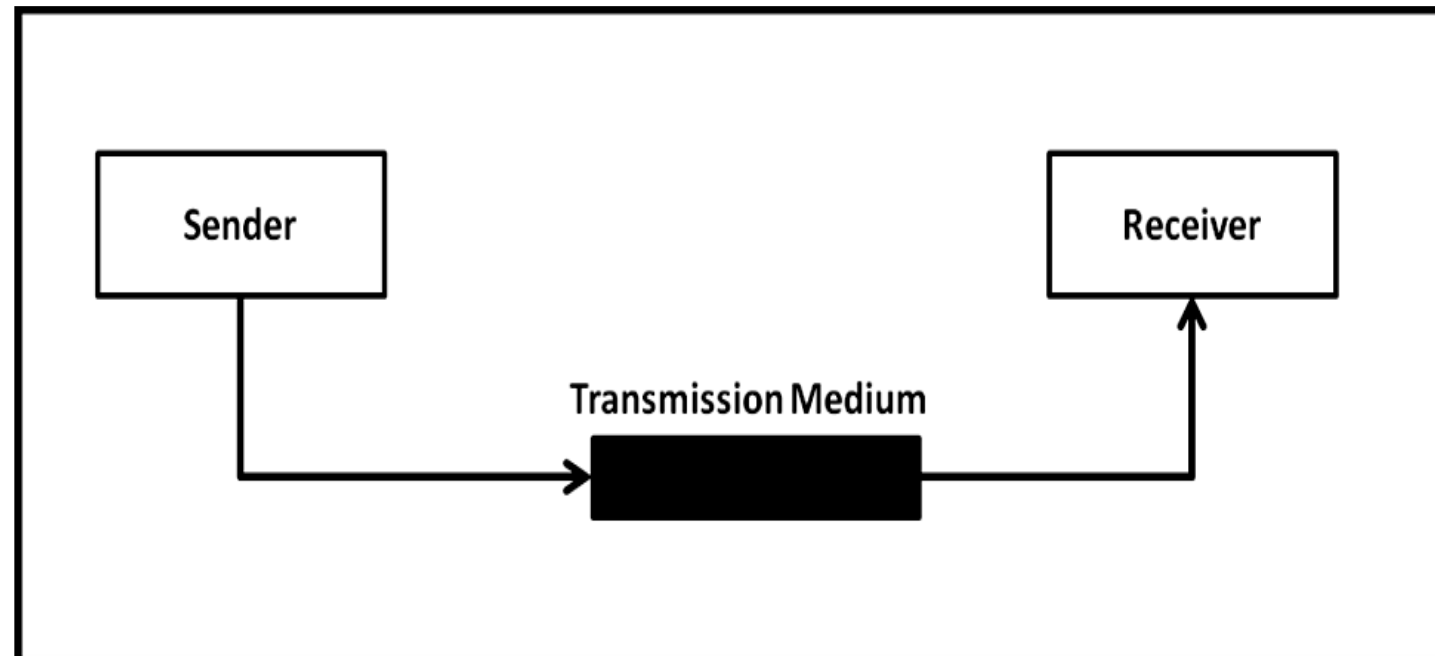
- Standardized architecture
- Modular design
- Easy troubleshooting

Limitations

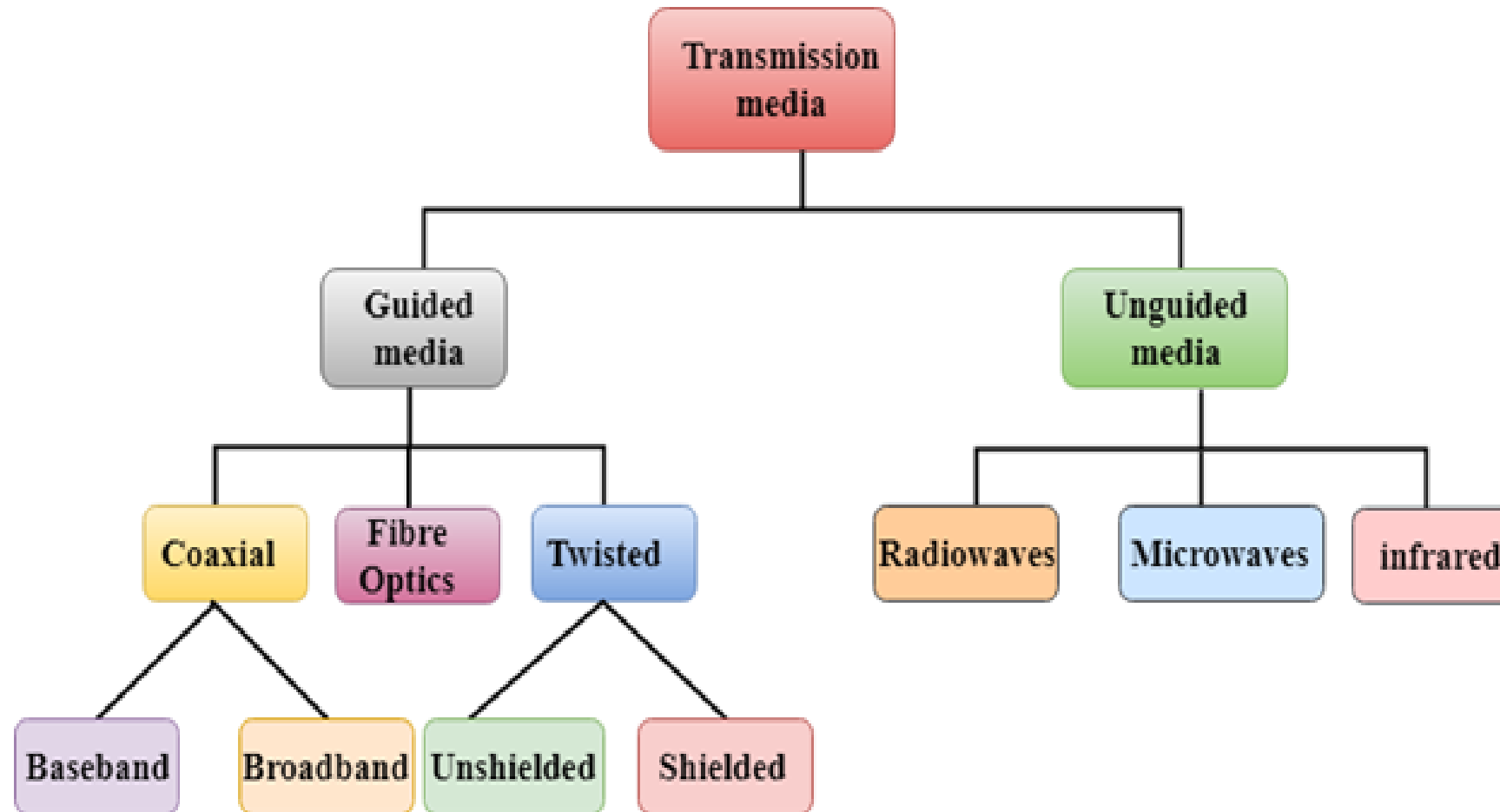
- Complex implementation
- Mostly conceptual

TRANSMISSION MEDIA:

- Transmission media is a means by which a communication **signal is carried** from one system to another.
- A transmission medium can be defined as anything that can carry information from a source to a destination.
- The transmission medium is usually free space, metallic cable or fiber-optic cable.



Categories of transmission media:



Transmission Media- Twisted Pair Cable, Coaxial Cable, Fibre-Optics Cable, Radio frequency Allocation, Terrestrial Microwave, Infrared rays, Satellite Communication, Cellular Telephony.

Guided Media:

- Guided Transmission media uses a **cabling** system that guides the data signals along a specific path.
- Guided media also known as **Bounded** media, which are those that provide a channel from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.
- Out of these twisted-pair cable, coaxial cable transport signals in the form of electric signals and fibre-optic cable transport signals in the form of light.

Types:

- 1. Twisted-Pair Cable**
- 2. Coaxial Cable**
- 3. Fibre-Optic Cable**

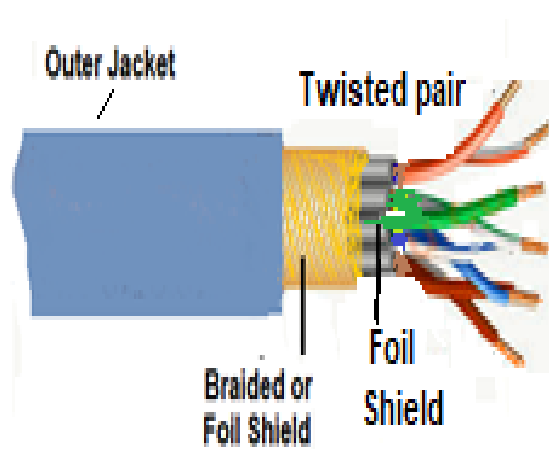
Twisted-pair cable:

- The 2 wires are twisted together in **pairs**.
- Each pair would consist of wire used for the **+ve data signal** and a wire used for the **-ve data signal**. Any noise that appears on +ve / -ve wire of the pair would occur on the other wire.
- Because the wires are opposite polarities, they are 180 degrees out of phase (180-degree phases) when the noise appears on both wires, it cancels or nulls itself out at the receiving end.
- Twisted pair cables are most effectively used in a system that uses a balanced line method of transmission.
- **Modulation** used by twisted pair cable is **Line coding** is the modulation of an electrical charge so that each side of a connection knows what is a one and what is a zero.

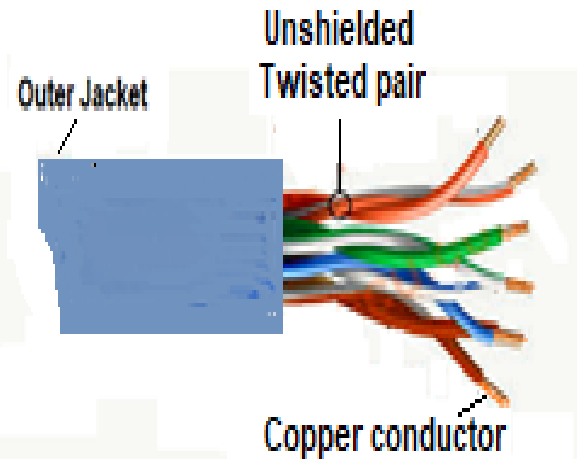


Types of Twisted Pair Cable :

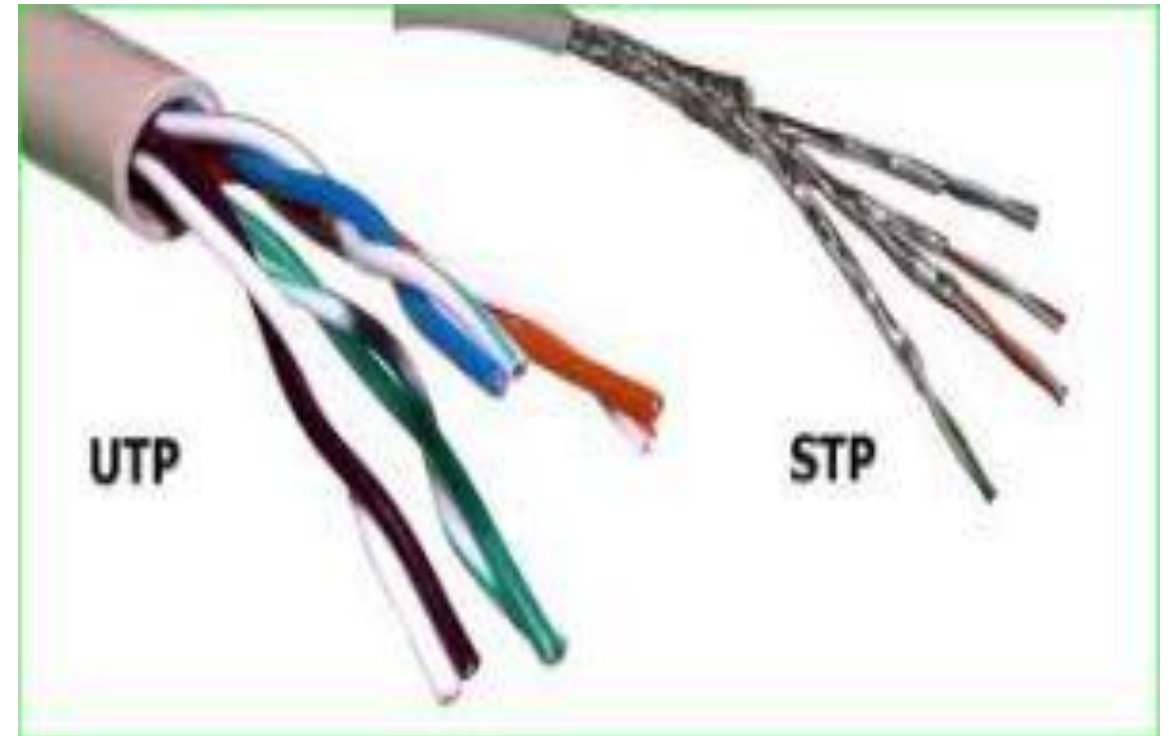
- Cables with the shield are called Shielded Twisted Pair and commonly abbreviated **STP**.
- Cables without a shield are called Unshielded Twisted Pair or **UTP**.
- UTP or unshielded twisted pair cable is used on Ethernet. UTP cable often is installed using a Registered Jack 45 (**RJ-45**) connector. The RJ-45 is an 8-wire connector (4 twisted pair cables) used commonly to connect computers onto a LAN, especially Ethernets.



Shielded Twisted
pair cable



Unshielded Twisted
pair cable



Examples:

- Telephone lines
- LAN
- Digital Subscriber Line (DSL)
 - Category 1 – UTP used in telephone lines with data rate < 0.1 Mbps
 - Category 2 – UTP used in transmission lines with a data rate of 2 Mbps
 - Category 3 – UTP used in LANs with a data rate of 10 Mbps
 - Category 4 – UTP used in Token Ring networks with a data rate of 20 Mbps
 - Category 5 – UTP used in LANs with a data rate of 100 Mbps
 - Category 6 – UTP used in LANs with a data rate of 200 Mbps
 - Category 7 – STP used in LANs with a data rate of 10 Mbps

Coaxial cable:

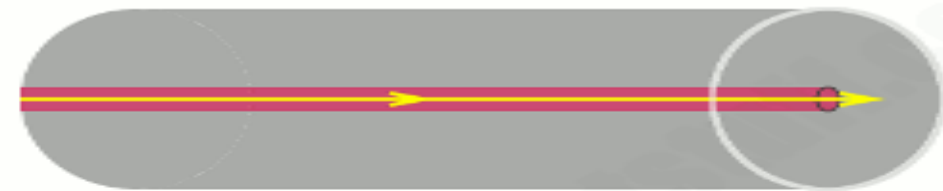
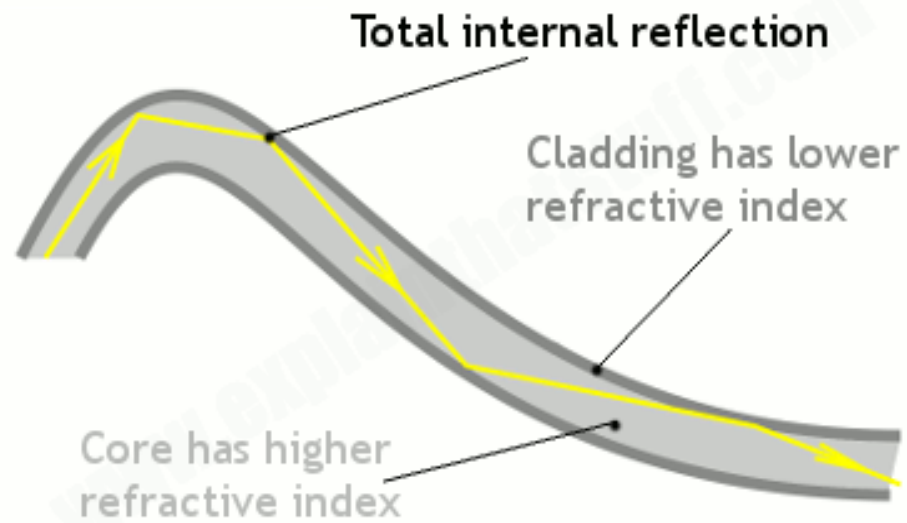
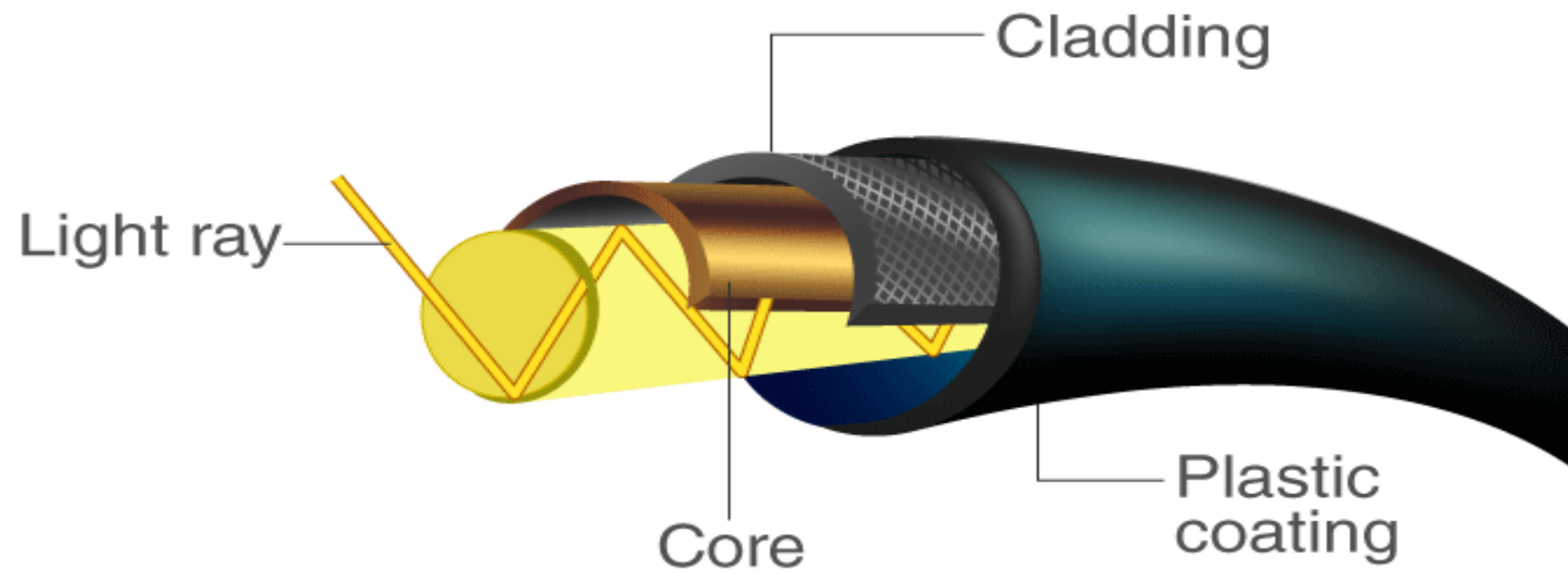
- Coaxial cable consists of 2 conductors. The inner conductor is contained inside the insulator with the other conductor weaves around it providing a shield.
- An insulating protective coating called a **jacket** covers the outer conductor.
- The outer shield protects the inner conductor from outside electrical signals.
- The distance between the outer conductor (Shield) and inner conductor plus the type of material used for insulating the inner conductor determine the cable properties or impedance (resistance).
- The excellent control of the impedance characteristics of the cable allow higher data rates to be transferred than twisted pair cable.
- The coaxial cable is a transmission line that is used to transport **electromagnetic signals** from source to the destination.
- Coaxial cables are used to connect the antennas to their transmitters and receivers.
- **Example:** we can consider the satellite based DTH **TV set** system. The receiving parabolic **dish antenna** (sender) is connected with a coaxial cable to transport the signals to the **set-top box** (receiver) of the **TV set**.

Coaxial cable



Fibre-Optics Cable:

- Optical fiber consists of **thin glass fiber** that can carry information at frequencies in the visible **light** spectrum.
- The typical optical fiber consists of a very narrow strand of glass called the **cladding**.
- Coating the cladding is a protective coating consisting of plastic, it is called the **jacket**.
- The device generating the message has it in electromagnetic form (**electrical signal**); this must be converted into light (i.e., **optical signal**) to send it on optic fiber cable. The process of converting light to electric signal is done on the receiving side.
- Whenever light gestures are transmitted through the optical cable, it moves forward within a **sequence of zigzag bounces**.
- **Transmitters:** The most used optical transmitters are semiconductor devices such as light-emitting diodes (**LEDs**) and laser diodes. Today, LEDs have been largely superseded by **VCSEL** (Vertical Cavity Surface Emitting **Laser**) devices, which offer improved speed.
- **Receivers:** The optical receiver is a photodetector which converts light into electricity.



Single-mode fiber



Multi-mode fiber

Advantages:

- 1. Small size and light weight:** The size of the optical fibers is very small. Therefore, many optical fibers can fit into a cable of small diameter.
- 2. Easy availability and low cost:** The material used for the manufacturing of optical fibers is —Silica glass. This material is easily available. So, the optical fibers cost lower than the cables with metallic conductors.
- 3. No electrical or electromagnetic interference:** Since the transmission takes place in the form of light rays the signal is not affected due to any electrical or electromagnetic Interference.
- 4. Large Bandwidth:** As the light rays have a very high frequency in GHz range, the bandwidth of the optical fiber is extremely large.
- 5. Other advantages:** - No cross talk inside the optical fiber cable. Signal can be sent up to 100 times faster.

Examples: cable television , Internet, Telephone, Military and Space Applications, Automotive Industry, Lighting And Decorations

Comparison Among Twisted Pair, Co-axial Cable and Fiber Optics cable

Characteristics	UTP	STP	Coaxial Cables	Fiber Optic Cables
Bandwidth	10 Mbps - 100 Mbps	10 Mbps - 100 Mbps	10 Mbps	100 Mbps -1 Gbps
Maximum cable segment	100 meters	100 meters	200 - 500 meters	2 k.m. - 100 k.m.
Interference rating	Poor	Better than UTP	Better than Twisted Pair Cable	Very good as compared to any other cable
Installation cost	Cheap	Costly than UTP	Costlier than twisted pair wires	Costliest to install
Bend radius	360 degrees / feet	360 degrees / feet	360 degrees / feet or 30 degrees / feet	30 degrees / feet
Security	Low	Low	Low	High



TCP/IP MODEL

- Transmission Control Protocol(TCP)/Internet Protocol(IP)
- Developed for the Internet.
- The layers in the TCP/IP protocol suite do not exactly match those in the OSI model.
- The original TCP/IP protocol suite was defined as having four layers:
 - host-to-network,
 - internet,
 - transport,
 - and application.
- However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application.

Characteristics

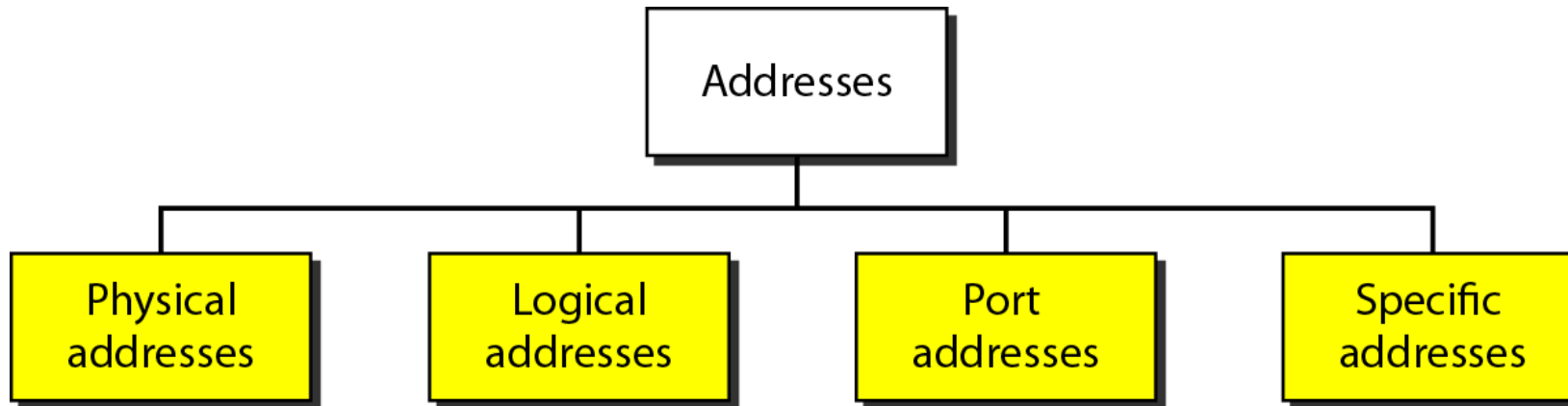
- Practical model
- Widely implemented
- Foundation of the Internet



ADDRESSING

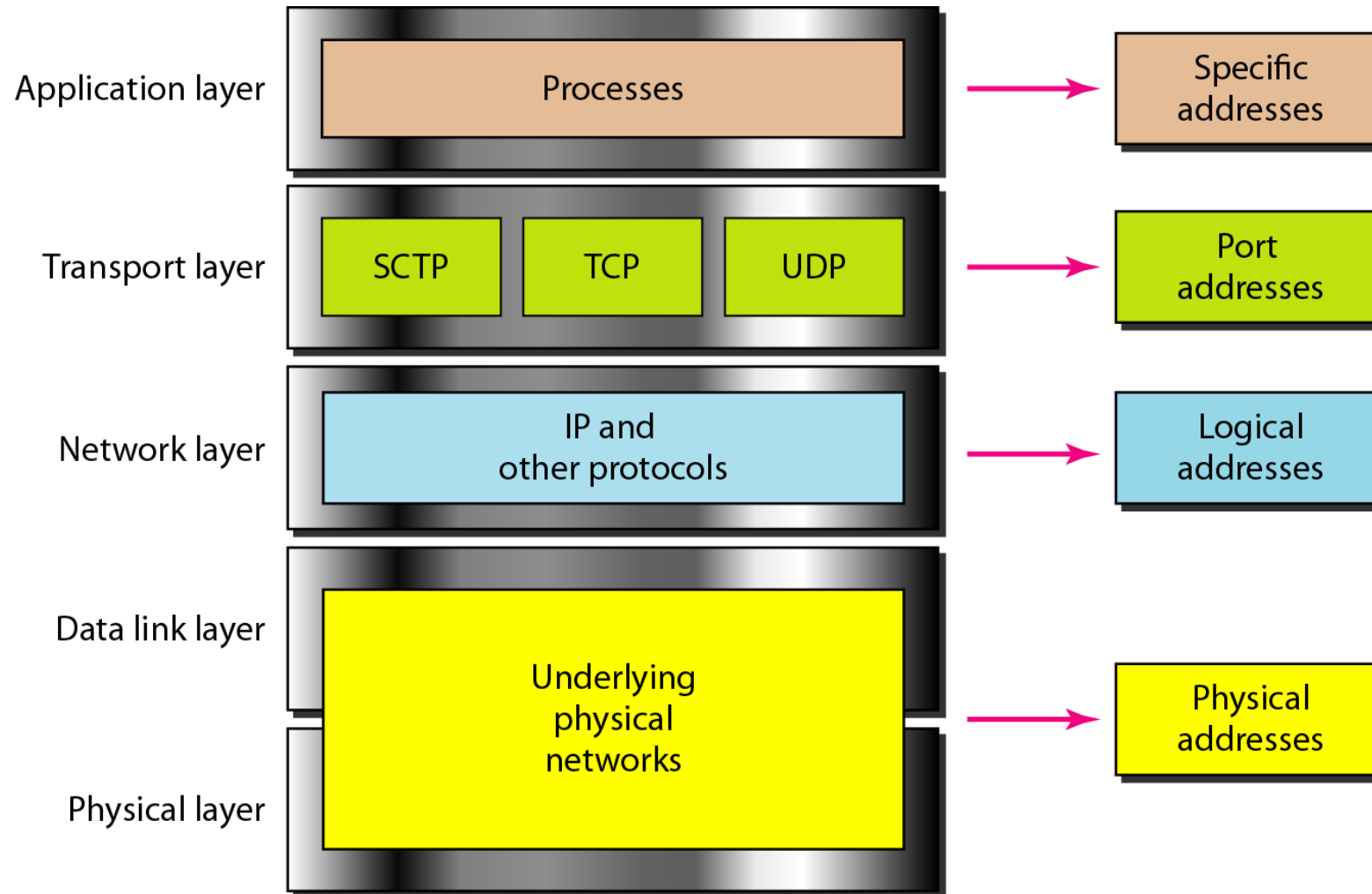
Four levels of addresses are used in an internet employing the TCP/IP protocols:

- physical
- logical
- port
- specific.



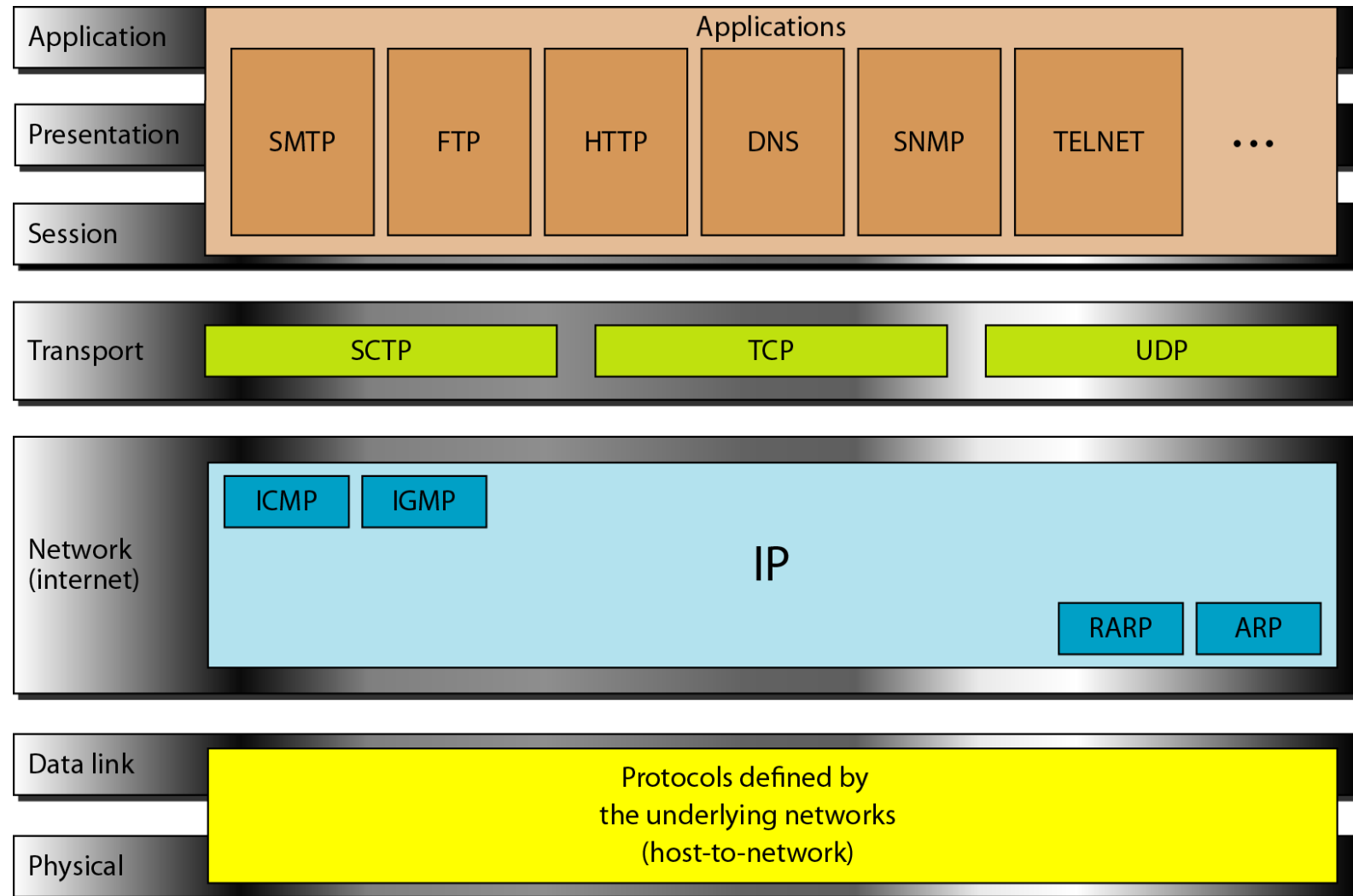


ADDRESSING



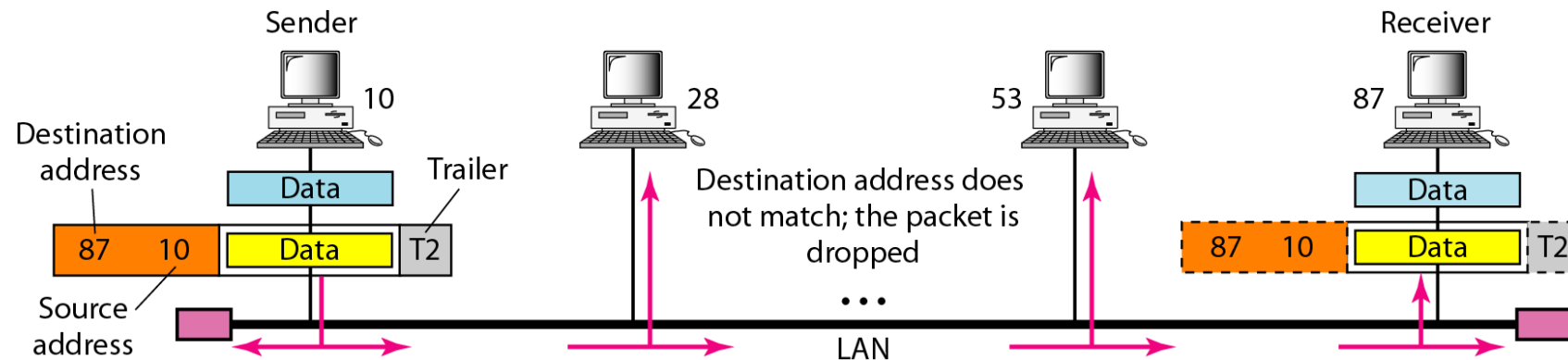


TCP/IP LAYERS



PHYSICAL ADDRESS

- A node with physical address 10 sends a frame to a node with physical address 87.
- The two nodes are connected by a link (bus topology LAN).
- As the figure shows, the computer with physical address 10 is the sender, and the computer with physical address 87 is the receiver.





PHYSICAL ADDRESS

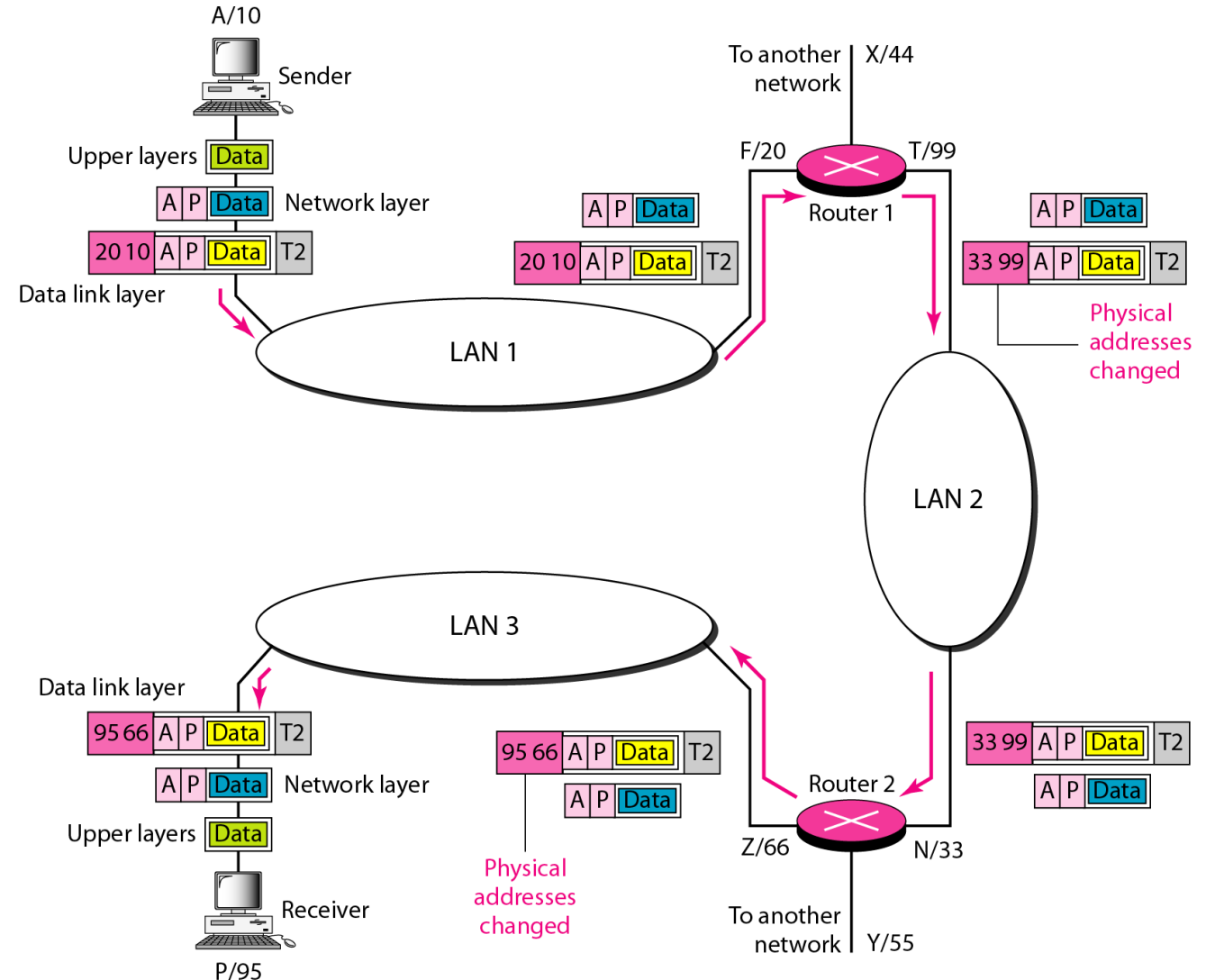
- *Most local-area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon, as shown below:*

07:01:02:01:2C:4B

A 6-byte (12 hexadecimal digits) physical address.

LOGICAL ADDRESS

- The figure shows a part of an internet with two routers connecting three LANs.
- Each device (computer or router) has a pair of addresses (logical and physical) for each connection.
- In this case, each computer is connected to only one link and therefore has only one pair of addresses.
- Each router, however, is connected to three networks (only two are shown in the figure).
- So each router has three pairs of addresses, one for each connection.





LOGICAL ADDRESS

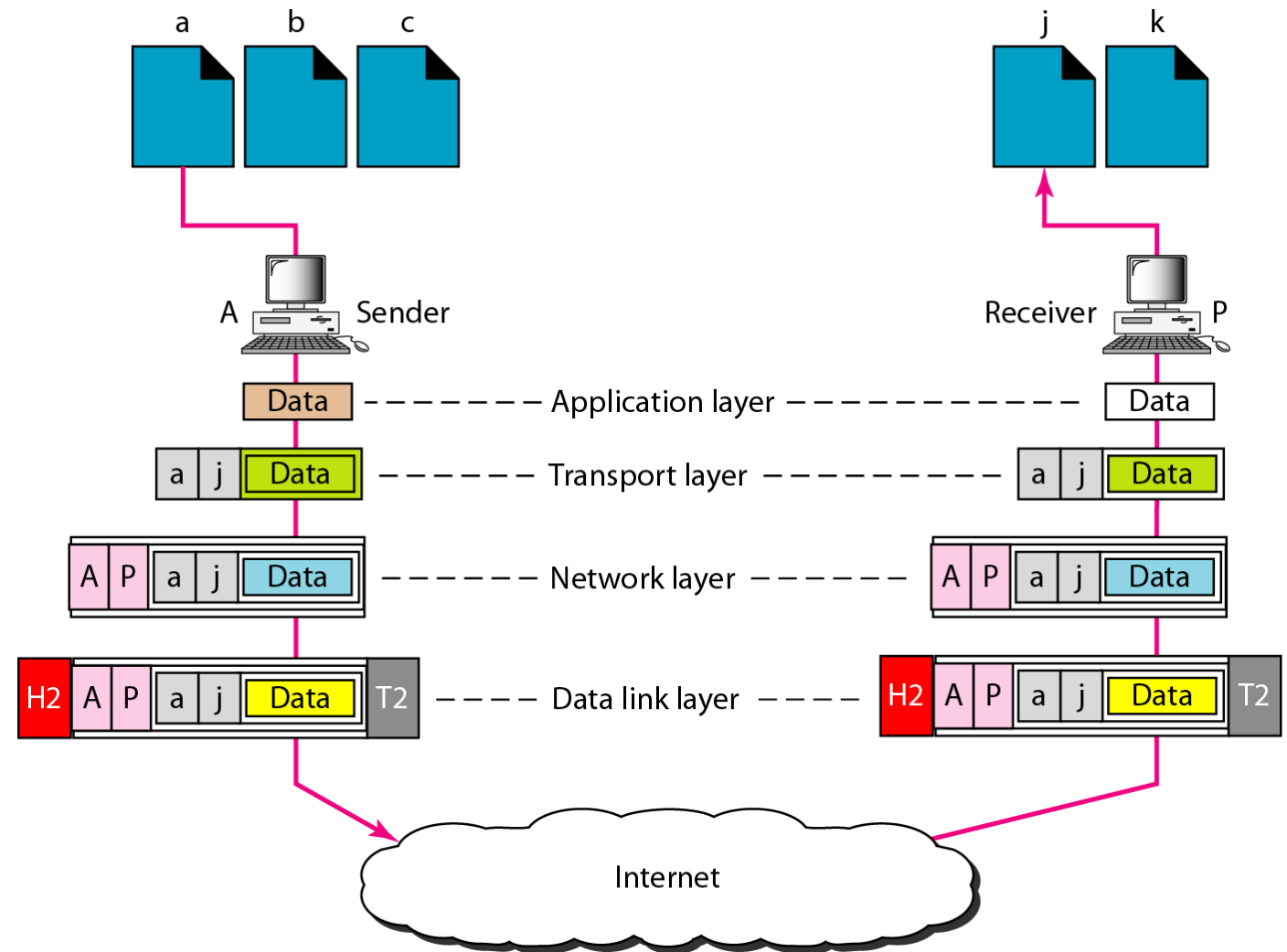
- An IP address is a 32-bit logical address (IPv4) represented by four decimal numbers separated by dots, used to uniquely identify a device on a network.*

192.168.1.10

A 32-bit logical address represented as four decimal numbers separated by dots.

PORT ADDRESS

- Figure shows two computers communicating via the Internet.
- The sending computer is running three processes at this time with port addresses *a*, *b*, and *c*.
- The receiving computer is running two processes at this time with port addresses *j* and *k*.
- Process *a* in the sending computer needs to communicate with process *j* in the receiving computer.
- Note that although physical addresses change from hop to hop, logical and port addresses remain the same from the source to destination.





PORT ADDRESS

- *A port address is a 16-bit address represented by one decimal number as shown.*

753

A 16-bit port address represented as one single number.



COMPARISON OF OSI AND TCP/IP MODELS

Feature	OSI Model	TCP/IP Model
Full Form	Open Systems Interconnection	Transmission Control Protocol/Internet Protocol
Developed By	International Organization for Standardization	U.S. Department of Defense
Number of Layers	7 Layers	4 Layers
Nature	Reference Model	Protocol Suite and Reference Model
Usage	Mainly for understanding and teaching networking concepts	Widely used in real-world networks and the Internet
Flexibility	More detailed and structured	Simpler and more practical
Examples	Conceptual framework	Internet, LANs, WANs



Thank you!

Doubts and Questions?